

UMC 0.11um RFCMOS AI Advanced Enhancement 3.3V SPICE Model Document

Version 0.5 phase 1

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1 Introduction

1.1 Revision history

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2010/7/29	Chih hung Chuang	1. Original release 2. Silicon-based model
0.2_p1	2010/10/21	Chih hung Chuang	Specify the geometry range
0.3_p1	2011/05/03	Laney Fan	1.Add noise figure in the models 2.Add the model of triple-well nmos transistor (copied from twin well nmos model).
0.4_p1	2011/08/16	Laney Fan	1. Modify the 3.3V Twin Well RF MOS model from 4 terminals to 6 terminals.
0.4_p2	2013/04/12	Laney Fan	1. Modify the verilog file for avoiding the noise as device is off.
0.5_p1	2014/01/28	Laney Fan	Correct the node names from "mosm d g s b" to "mosm 10 20 30 40"

1.2 General Information

Following Logic/MM model is used as the main body of the subckt RF MOSFET model and it would be fine tuned in order to improve Gm and Rout characteristics.

l110ae_33_rf_v051

More information about the model is specified below.

Model File Name:

Spectre: l110ae_33_rf_v051.lib.scs, l110ae_33_rf_v051.mdl.scs

Hspice: l110ae_33_rf_v051.lib, l110ae_33_rf_v051.mdl

Eldo: l110ae_33_rf_v051.lib.eldo, l110ae_33_rf_v051.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

Twin Well NMOS: n_33_rf (drain gate source body)

PMOS: p_33_rf (drain gate source body psub)

Triple Well NMOS: n_33bpw_rf (drain gate source body)

Corner Section:

tt : Typical n-ch MOSFET and Typical p-ch MOSFET

ff : Fast n-ch MOSFET and Fast p-ch MOSFET

ss : Slow n-ch MOSFET and Slow p-ch MOSFET

fnsp : Fast n-ch MOSFET and Slow p-ch MOSFET

snfp : Slow n-ch MOSFET and Fast p-ch MOSFET

Input Parameters:

lf : gate finger length in m

wf : gate finger width in m

nf : gate finger number

hfn_flag: local flag parameter for High-Frequency noise model, Default equals the global high-frequency noise flag parameter - hfnoise (Please refer to the chapter of “High Frequency Noise Model Enhancement” for detail)

mf: multiple factor for high frequency noise model (for Hspice and Eldo only). mf should be always equal to the instant multiple factor parameter “m”, in order to get accurate high frequency noise simulation results.

no_cnt: number of source/drain contact row. Default is 1.

p_area_nw: area of NWELL of a single device foe multiple devices in shared NWELL. Default value will be calculated based on Pcell. Unit is m^2 .

p_pj_nw: perimeter of NWELL of a single device foe multiple devices in shared NWELL. Default value will be calculated based on Pcell. Unit is m.

p_area_dnw: area of DWL of a single device foe multiple devices in shared NWELL. Default value will be calculated based on Pcell. Unit is m^2 .

(for Well Sharing in PMOS):

p_area_nw: area of NWELL of a single device for multiple devices in shared NWELL. Default value will be calculated based on Pcell. Unit is m^2 .

p_pj_nw: perimeter of NWELL of a single device for multiple devices in shared NWELL. Default value will be calculated based on Pcell. Unit is m.

p_area_dnw: area of DNW of a single device for multiple devices in shared NWELL. Default is $1e-15$. Unit is m^2 .

Geometry scaling(before 90% shrink):

NMOS:

Finger Length (lf): $0.34\mu m \sim 1\mu m$

Finger Width (wf): $3\mu m \sim 12\mu m$

Finger No (nf): $2 \sim 32$

Number of S/D contact row (con): 1

Support even and odd finger number

PMOS:

Finger Length (lf): $0.34\mu m \sim 1\mu m$

Finger Width (wf): $2.5\mu m \sim 10\mu m$

Finger No (nf): $2 \sim 32$

Number of S/D contact row (con): 1

Support even and odd finger number

Biasing coverage:

Voltage: Upto $1.1 \cdot V_{dd}$ ($V_{dd}=3.3V$)

Frequency: upto 20.1GHz

Temperature coverage:

$-55\text{ C} \sim +125\text{ C}$

Simulator Versions:

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 6.2.1.299

Mentor Graphics ELDO version 6.11_1.1

Wafer Information:

Table 1-2 Wafer Information

Mask ID	BSW0FA
Lot ID	D4YAA
Wafer Number	2

1.3 Model Usage:

1.3.1 For Cadence Spectre

- Copy ModelFileName.lib.scs and ModelFileName.mdl.scs & l110ae_33_rf_s.va into your simulation directory.
- define global high frequency noise model parameter (hfnoise: 1 or 0) in the input netlist (Please refer to the chapter of “High Frequency Noise Model Enhancement” for detail)
parameters hfnoise=0
- Add the following statement in the input netlist to the simulator
include "./ModelFileName.lib.scs " section = CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
- SetOption1 options scale=0.9
- Follow the example below to call the model
x1 (drain gate source body) n_33_rf lf=0.12e-6 wf=12e-6 nf=16 m=1
+hfn_flag=1

1.3.2 For Synopsis Hspice

- Copy ModelFileName.lib and ModelFileName.mdl & l110ae_33_rf_h.va into your simulation directory.
- define global high frequency noise model parameter (hfnoise: 1 or 0) in the input netlist (Please refer to the chapter of “High Frequency Noise Model Enhancement” for detail)
.param hfnoise=0
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_33_rf lf=0.12e-6 wf=12e-6 nf=16 m=1
+hfn_flag=1

1.3.3 For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo and ModelFileName.mdl.eldo & l110ae_33_rf_e.va into your simulation directory.
- define global high frequency noise model parameter (hfnoise: 1 or 0) in the input netlist (Please refer to the chapter of “High Frequency Noise Model Enhancement” for detail)
.param hfnoise=0
- Add the following statement in the input netlist to the simulator
.lib "/ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_33_rf lf=0.12e-6 wf=12e-6 nf=16 m=1
+hfn_flag=1

ModelFileName: l110ae_33_rf_v051

1.4 Device Architecture and Layout

1.4.1 Twin Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.14um

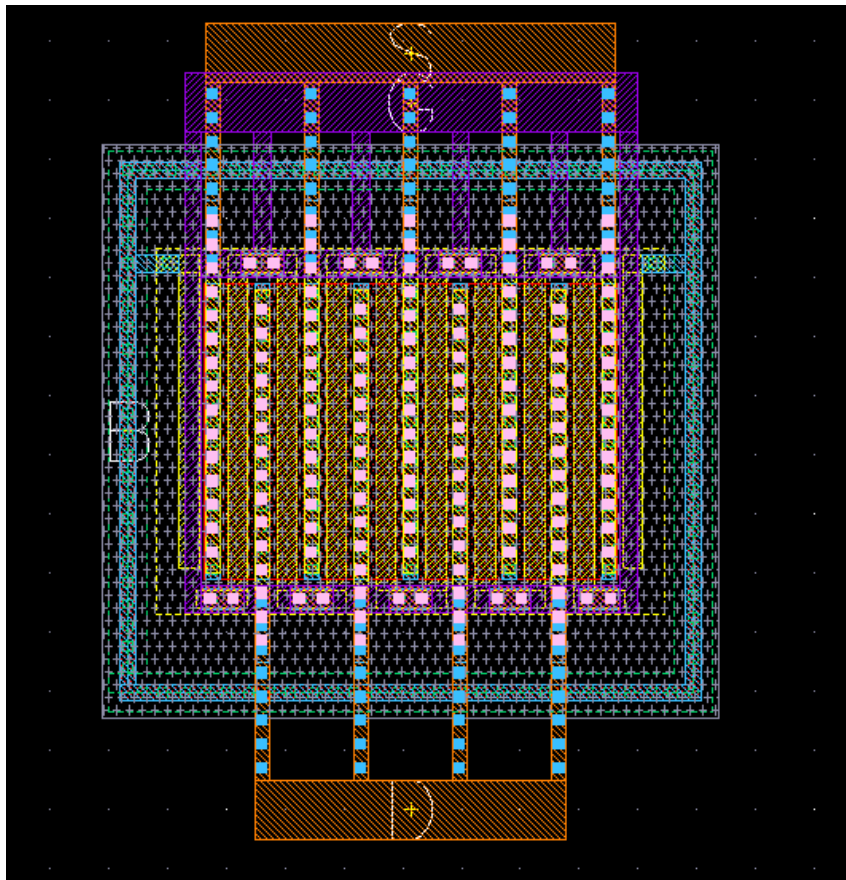


Figure 1-1 Example layout of Twin Well NMOS

Extension equivalent circuit model:

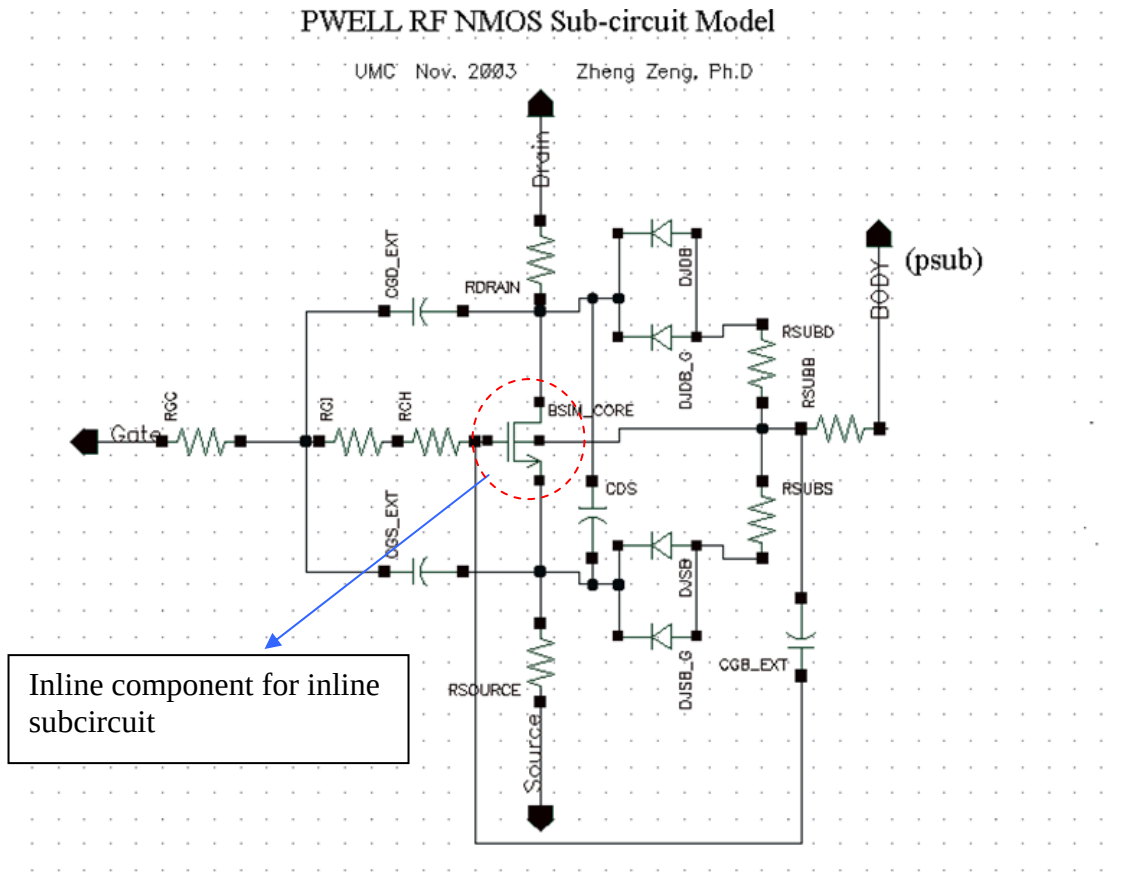


Figure 1-2 Extension equivalent circuit of TWIN-Well NMOS

1.4.2 PMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.14um

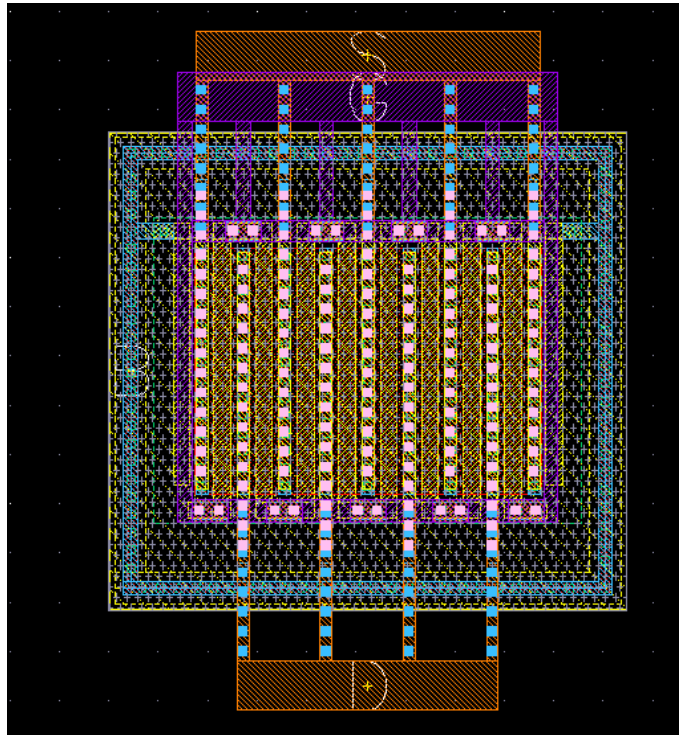


Figure 1-3 Example layout of PMOS

1.5 Model Features

BSIM3 :

The BSIM3 model in RF subcircuit model is based on Logic/MM model, with parameters refined to improve the fitting quality of Gm/ Rout and RF performance.

Device Usage General Guideline

- Due to the combined distributed effects in both the gate and channel at high frequency, the effective gate resistance (R_g) decreases when either gate length (l_f) or gate width (w_f) increases before reaching certain l_f & w_f , and then starts to increase as l_f or w_f continues to increase. If R_g is critical in your application, please carefully choose w_f , l_f and n_f to achieve lower R_g .
- Although model can support odd finger numbers, but it is recommended to use even finger number to reduce drain junction capacitance.
- Please use larger finger numbers if possible, since the process variation as well as the model fitting quality are better for larger finger number.

Note:

- Model can support odd finger numbers, but it is recommended to use even finger number to reduce drain junction capacitance.
- **DO NOT** use the model outside the above WF and LF range, since some binning parameters have been used in the BSIM3 model of RF subcircuit model to achieved better fitting quality, the BSIM3 model may fail outside the above WF, LF range.

2 RF Model Verification

2.1 Twin Well NMOS Model

2.1.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing condition for the parameters are:

ft : Cut-off frequency extracted @ Ft_max : Vd=3.3V, sweep Vg
vth : Linear Vth extracted @ Vd=0.1V @
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=3.3V using fixed current
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ -3.3V
Idsat : Drain current extracted @ Vd=Vg=3.3V
Idlin : Drain current extracted @ Vd=0.1V, Vg=3.3V
Ioff : Drain current extracted @ Vg=0V, Vd=3.3V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	12	12	6	3	12	12	6	3
	NF	16	16	32	16	16	16	32	16
SS	ft(GHz)	6.1836	23.252	22.505	20.674	NA	NA	NA	NA
	idsat(uA/um)	2.53E+02	4.93E+02	4.90E+02	4.90E+02	2.89E+02	5.36E+02	5.33E+02	5.28E+02
	vtlin(V)	7.62E-01	7.18E-01	7.28E-01	7.54E-01	7.06E-01	7.12E-01	7.14E-01	7.18E-01
	vtsat(V)	7.37E-01	5.56E-01	5.67E-01	5.93E-01	6.89E-01	5.54E-01	5.56E-01	5.61E-01
	body_effect(V)	1.02E+00	8.86E-01	8.86E-01	8.87E-01	5.39E-01	1.68E-01	1.68E-01	1.69E-01
	ioff(pA/um)	6.51E-01	1.43E+00	1.84E+00	2.87E+00	6.32E-02	4.54E-01	4.91E-01	5.71E-01
SNFP	ft(GHz)	6.5571	24.736	24.133	22.5	NA	NA	NA	NA
	idsat(uA/um)	2.75E+02	5.37E+02	5.35E+02	5.36E+02	3.12E+02	5.83E+02	5.80E+02	5.76E+02
	vtlin(V)	7.31E-01	6.73E-01	6.83E-01	7.08E-01	6.75E-01	6.69E-01	6.71E-01	6.74E-01
	vtsat(V)	7.07E-01	5.17E-01	5.27E-01	5.53E-01	6.59E-01	5.15E-01	5.17E-01	5.21E-01
	body_effect(V)	9.99E-01	8.67E-01	8.68E-01	8.69E-01	5.29E-01	1.65E-01	1.65E-01	1.66E-01
	ioff(pA/um)	6.53E-01	2.79E+00	2.88E+00	3.38E+00	6.58E-02	1.19E+00	1.19E+00	1.22E+00
TT	ft(GHz)	7.1391	27.814	27.344	25.805	NA	NA	NA	NA
	idsat(uA/um)	2.94E+02	5.79E+02	5.76E+02	5.77E+02	3.34E+02	6.30E+02	6.28E+02	6.24E+02
	vtlin(V)	7.05E-01	6.31E-01	6.42E-01	6.67E-01	6.48E-01	6.26E-01	6.28E-01	6.31E-01
	vtsat(V)	6.80E-01	4.58E-01	4.69E-01	4.95E-01	6.31E-01	4.64E-01	4.66E-01	4.69E-01
	body_effect(V)	9.99E-01	8.58E-01	8.59E-01	8.60E-01	5.27E-01	1.49E-01	1.49E-01	1.51E-01
	ioff(pA/um)	6.56E-01	1.16E+01	9.43E+00	6.52E+00	7.20E-02	5.08E+00	4.93E+00	4.68E+00
FNFP	ft(GHz)	7.7136	31.126	30.695	29.096	NA	NA	NA	NA
	idsat(uA/um)	3.14E+02	6.22E+02	6.19E+02	6.19E+02	3.57E+02	6.78E+02	6.76E+02	6.72E+02
	vtlin(V)	6.79E-01	5.89E-01	6.00E-01	6.25E-01	6.21E-01	5.83E-01	5.84E-01	5.87E-01
	vtsat(V)	6.54E-01	3.96E-01	4.07E-01	4.33E-01	6.04E-01	4.12E-01	4.13E-01	4.16E-01
	body_effect(V)	9.99E-01	8.49E-01	8.50E-01	8.51E-01	5.25E-01	1.33E-01	1.33E-01	1.35E-01
	ioff(pA/um)	6.64E-01	6.19E+01	4.65E+01	2.42E+01	8.55E-02	2.28E+01	2.21E+01	2.09E+01
FF	ft(GHz)	8.018	32.535	32.236	30.865	NA	NA	NA	NA
	idsat(uA/um)	3.35E+02	6.66E+02	6.63E+02	6.65E+02	3.79E+02	7.25E+02	7.22E+02	7.19E+02
	vtlin(V)	6.49E-01	5.46E-01	5.56E-01	5.81E-01	5.91E-01	5.40E-01	5.40E-01	5.42E-01
	vtsat(V)	6.24E-01	3.59E-01	3.70E-01	3.95E-01	5.74E-01	3.73E-01	3.74E-01	3.76E-01
	body_effect(V)	9.79E-01	8.30E-01	8.31E-01	8.33E-01	5.15E-01	1.30E-01	1.31E-01	1.32E-01
	ioff(pA/um)	6.79E-01	1.61E+02	1.21E+02	6.09E+01	1.17E-01	6.72E+01	6.57E+01	6.33E+01

Table 2-1 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	12	12	6	3	12	12	6	3
	NF	16	16	32	16	16	16	32	16
SS	ft(GHz)	4.3565	19.274	18.631	17.078	NA	NA	NA	NA
	idsat(uA/um)	1.81E+02	4.25E+02	4.22E+02	4.21E+02	2.24E+02	4.52E+02	4.50E+02	4.46E+02
	vtlin(V)	6.51E-01	5.97E-01	6.08E-01	6.33E-01	6.16E-01	6.33E-01	6.35E-01	6.40E-01
	vtsat(V)	6.23E-01	4.28E-01	4.40E-01	4.66E-01	5.94E-01	4.67E-01	4.69E-01	4.74E-01
	body_effect(V)	1.07E+00	9.25E-01	9.26E-01	9.28E-01	5.67E-01	1.74E-01	1.75E-01	1.76E-01
	ioff(pA/um)	2.27E+00	2.77E+02	2.22E+02	1.29E+02	1.94E+00	9.18E+01	8.74E+01	7.97E+01
SNFP	ft(GHz)	4.5933	20.376	19.853	18.471	NA	NA	NA	NA
	idsat(uA/um)	1.96E+02	4.61E+02	4.58E+02	4.58E+02	2.41E+02	4.90E+02	4.88E+02	4.84E+02
	vtlin(V)	6.19E-01	5.51E-01	5.61E-01	5.86E-01	5.84E-01	5.89E-01	5.91E-01	5.95E-01
	vtsat(V)	5.91E-01	3.88E-01	3.99E-01	4.24E-01	5.63E-01	4.28E-01	4.30E-01	4.34E-01
	body_effect(V)	1.05E+00	9.07E-01	9.08E-01	9.10E-01	5.57E-01	1.72E-01	1.72E-01	1.73E-01
	ioff(pA/um)	3.58E+00	6.06E+02	4.90E+02	2.87E+02	3.51E+00	2.04E+02	1.96E+02	1.82E+02
TT	ft(GHz)	5.0033	22.963	22.551	21.253	NA	NA	NA	NA
	idsat(uA/um)	2.10E+02	4.98E+02	4.95E+02	4.95E+02	2.57E+02	5.29E+02	5.27E+02	5.24E+02
	vtlin(V)	5.91E-01	5.07E-01	5.18E-01	5.43E-01	5.57E-01	5.46E-01	5.48E-01	5.51E-01
	vtsat(V)	5.63E-01	3.27E-01	3.38E-01	3.64E-01	5.36E-01	3.76E-01	3.78E-01	3.81E-01
	body_effect(V)	1.05E+00	8.98E-01	8.99E-01	9.01E-01	5.55E-01	1.55E-01	1.55E-01	1.57E-01
	ioff(pA/um)	5.81E+00	2.11E+03	1.69E+03	9.86E+02	6.32E+00	6.21E+02	6.00E+02	5.65E+02
FNFP	ft(GHz)	5.4119	25.765	25.387	24.05	NA	NA	NA	NA
	idsat(uA/um)	2.25E+02	5.37E+02	5.34E+02	5.33E+02	2.74E+02	5.69E+02	5.67E+02	5.64E+02
	vtlin(V)	5.63E-01	4.63E-01	4.73E-01	4.99E-01	5.29E-01	5.02E-01	5.04E-01	5.06E-01
	vtsat(V)	5.36E-01	2.63E-01	2.74E-01	3.00E-01	5.08E-01	3.24E-01	3.25E-01	3.28E-01
	body_effect(V)	1.05E+00	8.89E-01	8.90E-01	8.92E-01	5.53E-01	1.38E-01	1.39E-01	1.40E-01
	ioff(pA/um)	9.77E+00	7.74E+03	6.19E+03	3.61E+03	1.14E+01	1.90E+03	1.85E+03	1.76E+03
FF	ft(GHz)	5.5976	26.792	26.522	25.381	NA	NA	NA	NA
	idsat(uA/um)	2.39E+02	5.73E+02	5.70E+02	5.70E+02	2.90E+02	6.06E+02	6.04E+02	6.02E+02
	vtlin(V)	5.33E-01	4.19E-01	4.30E-01	4.54E-01	4.98E-01	4.58E-01	4.59E-01	4.61E-01
	vtsat(V)	5.06E-01	2.25E-01	2.36E-01	2.62E-01	4.78E-01	2.85E-01	2.86E-01	2.87E-01
	body_effect(V)	1.03E+00	8.70E-01	8.72E-01	8.74E-01	5.43E-01	1.36E-01	1.36E-01	1.38E-01
	ioff(pA/um)	1.70E+01	1.59E+04	1.29E+04	7.66E+03	2.12E+01	4.23E+03	4.15E+03	4.02E+03

Table 2-2 RF & Logic Corner Table at 125 °C

Temp=-55C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	12	12	6	3	12	12	6	3
	NF	16	16	32	16	16	16	32	16
SS	ft(GHz)	9.1099	27.067	26.232	24.105	NA	NA	NA	NA
	idsat(uA/um)	3.80E+02	5.57E+02	5.55E+02	5.56E+02	3.58E+02	5.92E+02	5.88E+02	5.83E+02
	vtlin(V)	8.53E-01	8.20E-01	8.30E-01	8.56E-01	7.84E-01	7.81E-01	7.83E-01	7.88E-01
	vtsat(V)	8.30E-01	6.65E-01	6.76E-01	7.02E-01	7.70E-01	6.30E-01	6.32E-01	6.36E-01
	body_effect(V)	9.65E-01	8.48E-01	8.48E-01	8.47E-01	5.17E-01	1.61E-01	1.61E-01	1.63E-01
	ioff(pA/um)	6.49E-01	6.50E-01	1.26E+00	2.60E+00	6.09E-02	6.12E-02	1.22E-01	2.44E-01
SNFP	ft(GHz)	9.7643	28.975	28.306	26.392	NA	NA	NA	NA
	idsat(uA/um)	4.15E+02	6.10E+02	6.09E+02	6.11E+02	3.88E+02	6.46E+02	6.42E+02	6.38E+02
	vtlin(V)	8.23E-01	7.76E-01	7.86E-01	8.11E-01	7.54E-01	7.39E-01	7.41E-01	7.44E-01
	vtsat(V)	8.00E-01	6.27E-01	6.37E-01	6.62E-01	7.40E-01	5.91E-01	5.93E-01	5.97E-01
	body_effect(V)	9.45E-01	8.29E-01	8.29E-01	8.28E-01	5.06E-01	1.58E-01	1.59E-01	1.60E-01
	ioff(pA/um)	6.49E-01	6.50E-01	1.26E+00	2.60E+00	6.09E-02	6.18E-02	1.23E-01	2.45E-01
TT	ft(GHz)	10.589	32.474	31.962	30.148	NA	NA	NA	NA
	idsat(uA/um)	4.41E+02	6.54E+02	6.52E+02	6.55E+02	4.15E+02	6.97E+02	6.94E+02	6.90E+02
	vtlin(V)	7.98E-01	7.37E-01	7.47E-01	7.72E-01	7.27E-01	6.97E-01	6.98E-01	7.01E-01
	vtsat(V)	7.75E-01	5.70E-01	5.81E-01	6.07E-01	7.13E-01	5.41E-01	5.42E-01	5.45E-01
	body_effect(V)	9.45E-01	8.20E-01	8.21E-01	8.20E-01	5.05E-01	1.43E-01	1.43E-01	1.45E-01
	ioff(pA/um)	6.49E-01	6.55E-01	1.26E+00	2.60E+00	6.09E-02	6.58E-02	1.26E-01	2.48E-01
FNFP	ft(GHz)	11.389	36.207	35.742	33.845	NA	NA	NA	NA
	idsat(uA/um)	4.67E+02	6.99E+02	6.97E+02	6.99E+02	4.44E+02	7.49E+02	7.46E+02	7.42E+02
	vtlin(V)	7.73E-01	6.96E-01	7.06E-01	7.32E-01	7.01E-01	6.54E-01	6.55E-01	6.58E-01
	vtsat(V)	7.50E-01	5.10E-01	5.21E-01	5.47E-01	6.86E-01	4.89E-01	4.90E-01	4.93E-01
	body_effect(V)	9.44E-01	8.11E-01	8.12E-01	8.12E-01	5.03E-01	1.27E-01	1.27E-01	1.29E-01
	ioff(pA/um)	6.49E-01	7.07E-01	1.30E+00	2.61E+00	6.09E-02	9.70E-02	1.56E-01	2.76E-01
FF	ft(GHz)	11.953	38.042	37.733	36.084	NA	NA	NA	NA
	idsat(uA/um)	5.01E+02	7.52E+02	7.50E+02	7.53E+02	4.72E+02	8.01E+02	7.98E+02	7.95E+02
	vtlin(V)	7.45E-01	6.54E-01	6.64E-01	6.89E-01	6.71E-01	6.11E-01	6.12E-01	6.13E-01
	vtsat(V)	7.22E-01	4.74E-01	4.85E-01	5.11E-01	6.57E-01	4.51E-01	4.51E-01	4.53E-01
	body_effect(V)	9.25E-01	7.93E-01	7.94E-01	7.93E-01	4.93E-01	1.24E-01	1.25E-01	1.27E-01
	ioff(pA/um)	6.49E-01	8.58E-01	1.40E+00	2.65E+00	6.09E-02	2.17E-01	2.74E-01	3.89E-01

Table 2-3 RF & Logic Corner Table at -55 °C

2.1.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device					Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
NFET751	0.34	3	16	4	0.668	0.672	0.494	0.5	556.18	568.35	6	6.501	24.367	24.873
NFET752	0.34	6	16	2	0.661	0.647	0.481	0.473	535.54	562.6	47.96	9.458	26.276	25.832
NFET753	0.34	12	16	1	0.656	0.637	0.476	0.463	508.04	555.74	13.08	11.607	26.756	26.179
NFET754	0.34	2	16	5	0.675	0.692	0.504	0.52	565.19	573.7	47.81	6.058	22.593	23.615
NFET755	0.34	6	4	8	0.671	0.647	0.49	0.473	548.88	568.59	67.92	9.687	25.615	24.478
NFET757	0.34	6	32	1	0.658	0.647	0.477	0.473	529.17	558.84	360.69	9.42	26.27	25.969
NFET760	0.34	3	8	8	0.671	0.672	0.494	0.5	560.07	570.09	73.81	6.654	24.137	24.447
NFET761	0.34	3	11	6	0.671	0.672	0.494	0.5	557.34	569.31	9.47	6.57	24.265	24.713
NFET762	0.34	3	32	2	0.669	0.672	0.492	0.5	553.1	566.43	59.03	6.425	24.585	25.076
NFET764	0.5	9	16	1	0.713	0.682	0.651	0.628	419.57	438.17	-20.18	0.928	17.085	16.694
NFET765	0.7	12	16	1	0.703	0.698	0.663	0.668	341.33	358.11	5.13	0.664	11.092	11.088
NFET766	1	12	16	1	0.692	0.707	0.656	0.682	281.06	287.74	67.49	0.656	7.023	6.968
NFET768	0.34	12	4	4	0.668	0.636	0.489	0.463	525.36	566.64	53.8	11.721	26.404	24.885
NFET770	0.34	12	32	1	0.652	0.637	0.472	0.463	481.66	545.45	10.31	11.587	26.439	25.943
NFET771	0.42	12	4	4	0.711	0.664	0.62	0.577	459.91	492.54	19.33	1.096	20.724	19.74
NFET772	0.5	12	11	2	0.713	0.68	0.652	0.627	403.1	436.32	-12.05	0.731	16.726	16.645
NFET773	0.7	12	32	1	0.701	0.698	0.661	0.668	330.79	353.7	5.67	0.644	10.916	11.072
NFET774	0.34	3	4	1	0.68	0.672	0.497	0.5	576.87	573.58	1274.07	6.959	22.733	23.676

Table 2-4 Summary Table for Twin Well RF NMOS

Measure: Measurement result.

Model: Simulation result.

2.1.3 Simulation results for WF, LF and NF trends

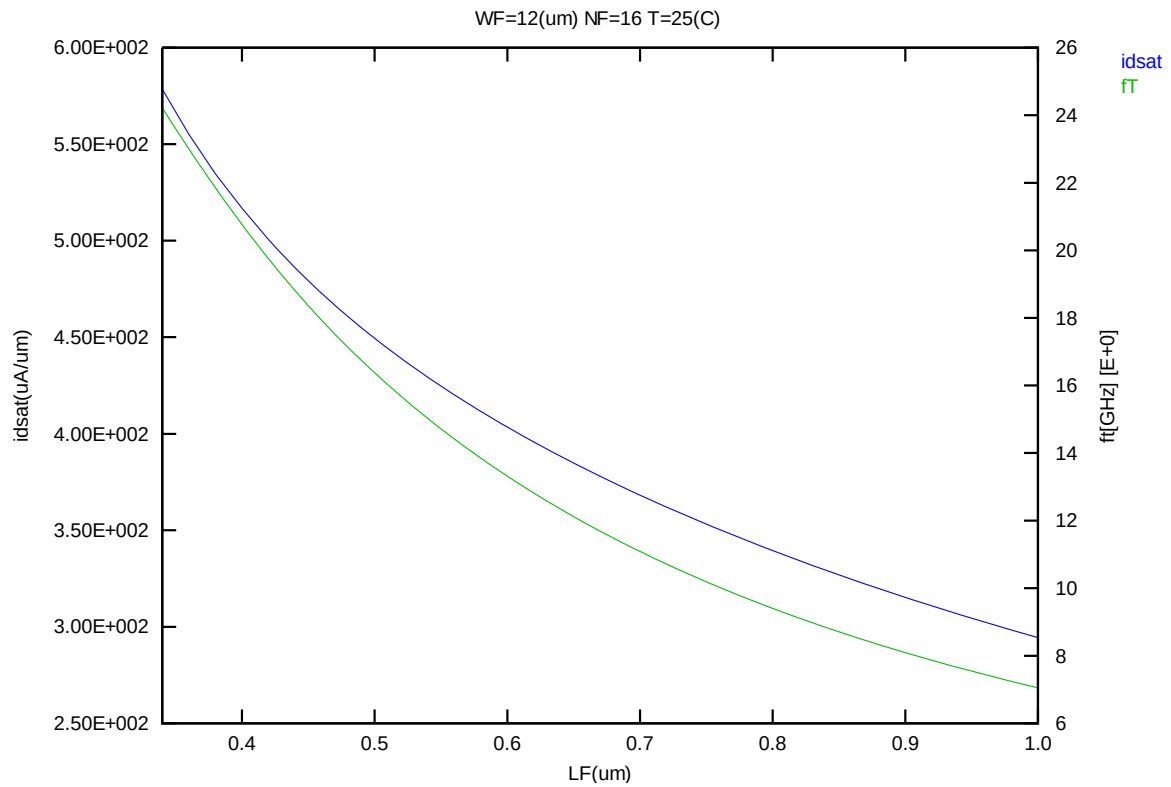


Figure 2-1 fT and $idsat$ vs. channel length

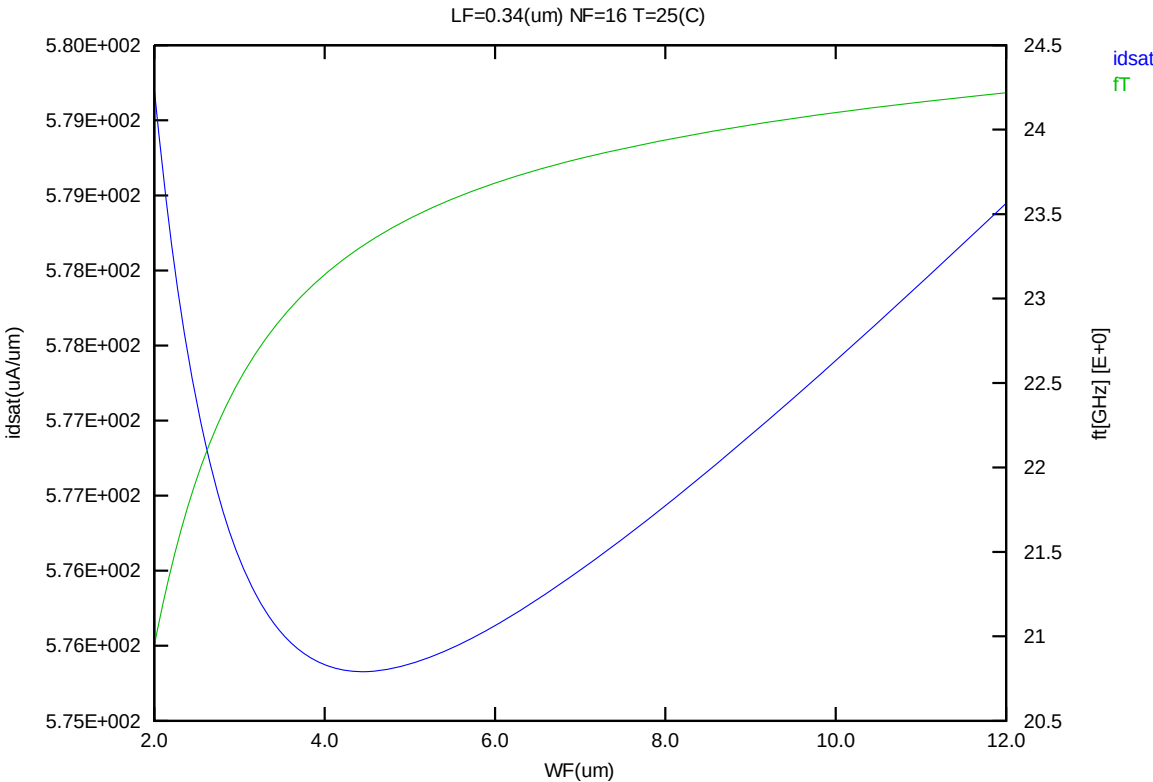


Figure 2-2 ft and idsat vs. channel width

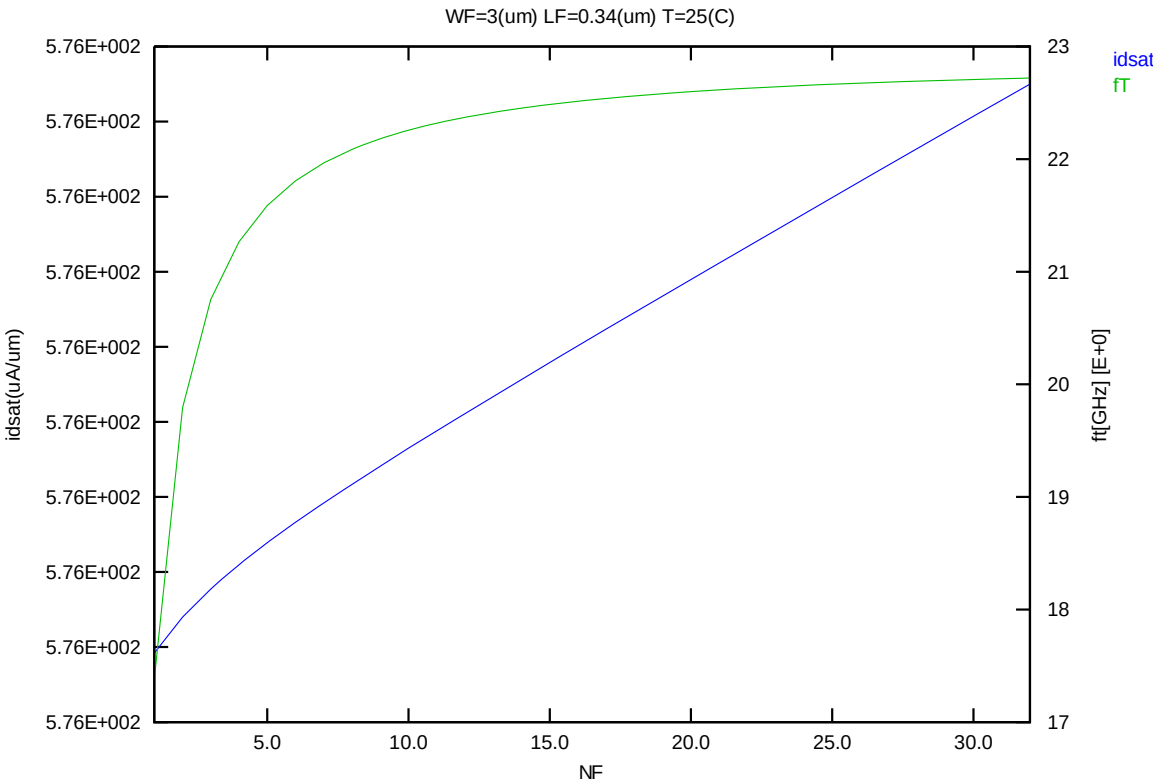


Figure 2-3 ft and idsat vs. finger number

2.1.4 DC & RF Fitting curves for individual devices

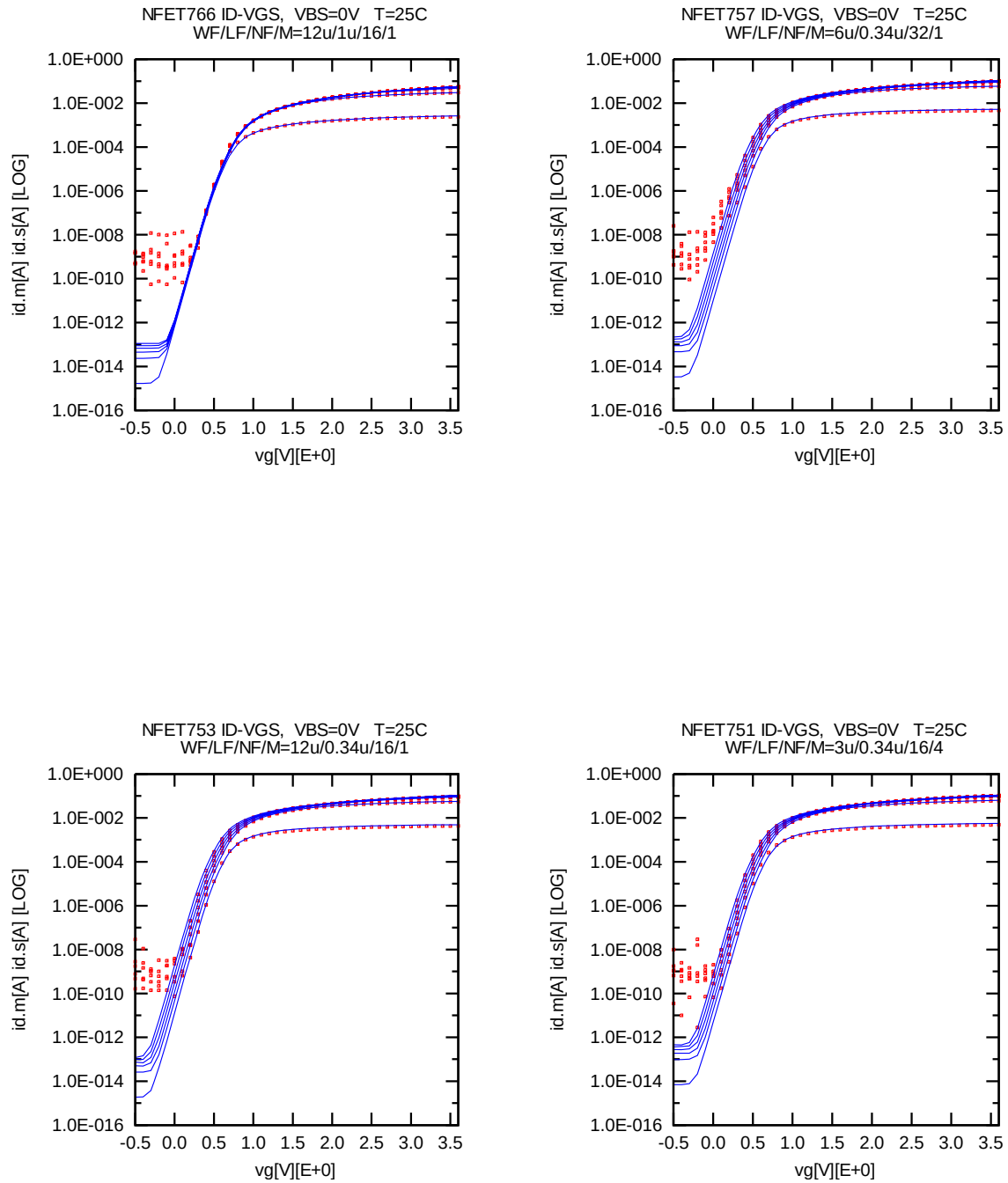
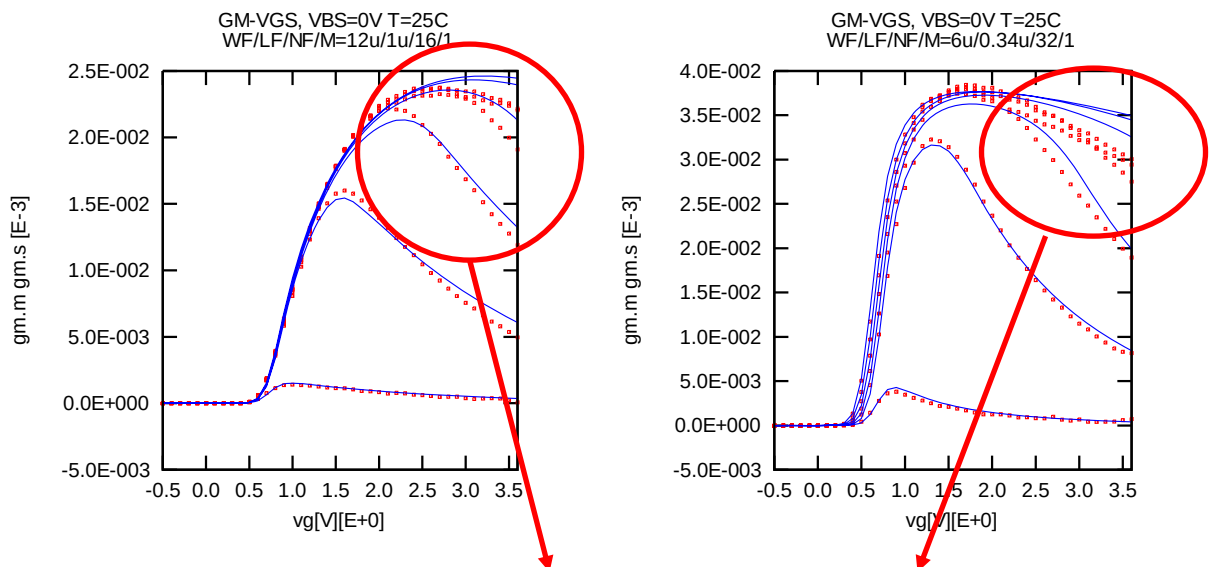


Figure 2-4 Id-Vg (25°C)



RF FET need large enough total gate width in order to get smooth and reliable S parameter measurement. However, too large gate width will cause self heating effect at high Vd and Vg bias, as the negative output resistance seen in the plots.

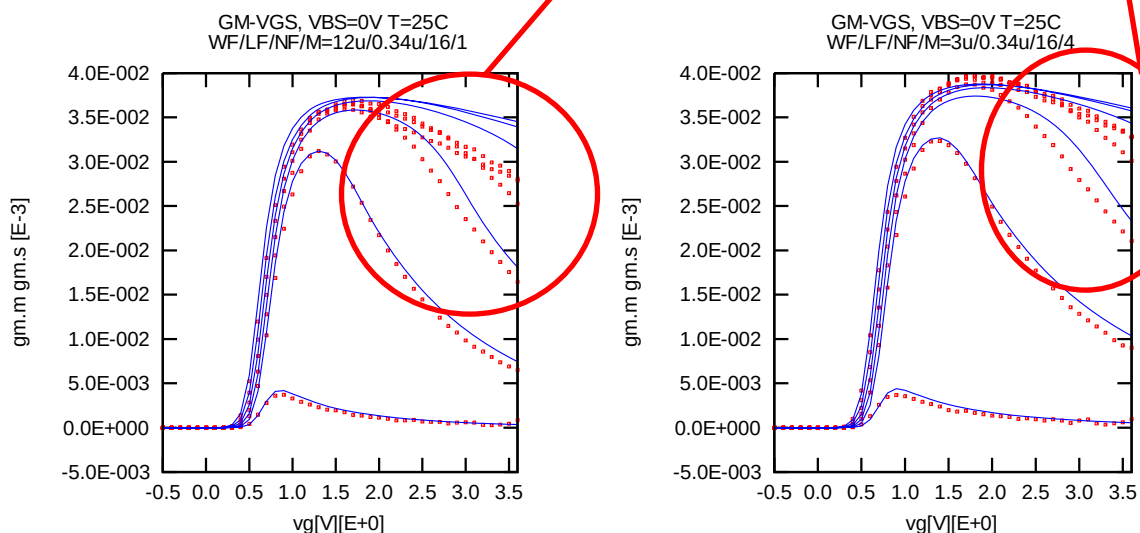
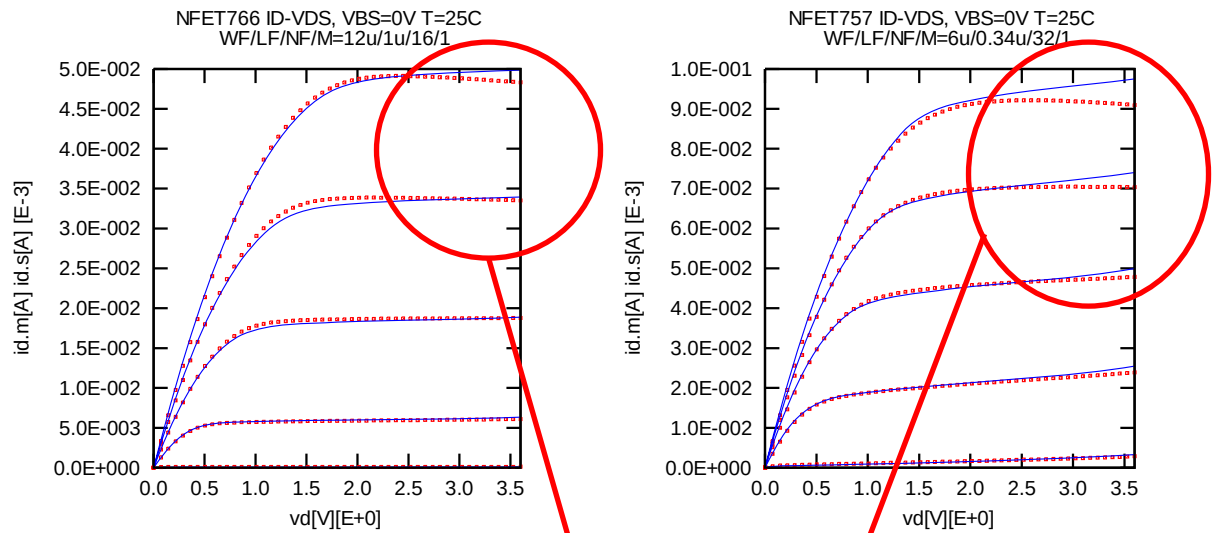


Figure 2-5 Gm-Vg (25°C)



RF FET need large enough total gate width in order to get smooth and reliable S parameter measurement. However, too large gate width will cause self heating effect at high Vd and Vg bias, as the negative output resistance seen in the plots.

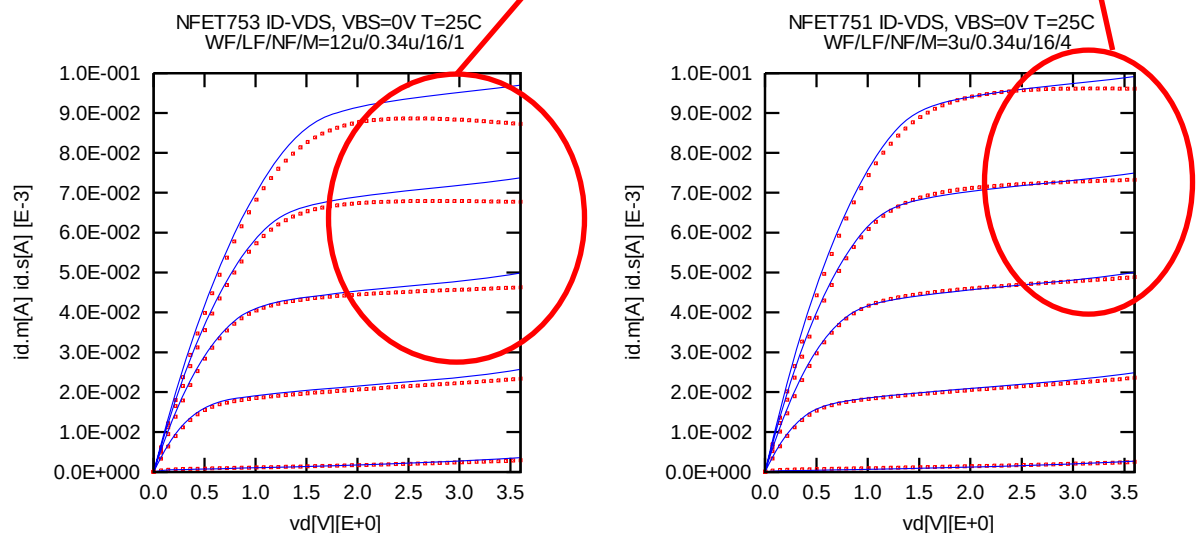
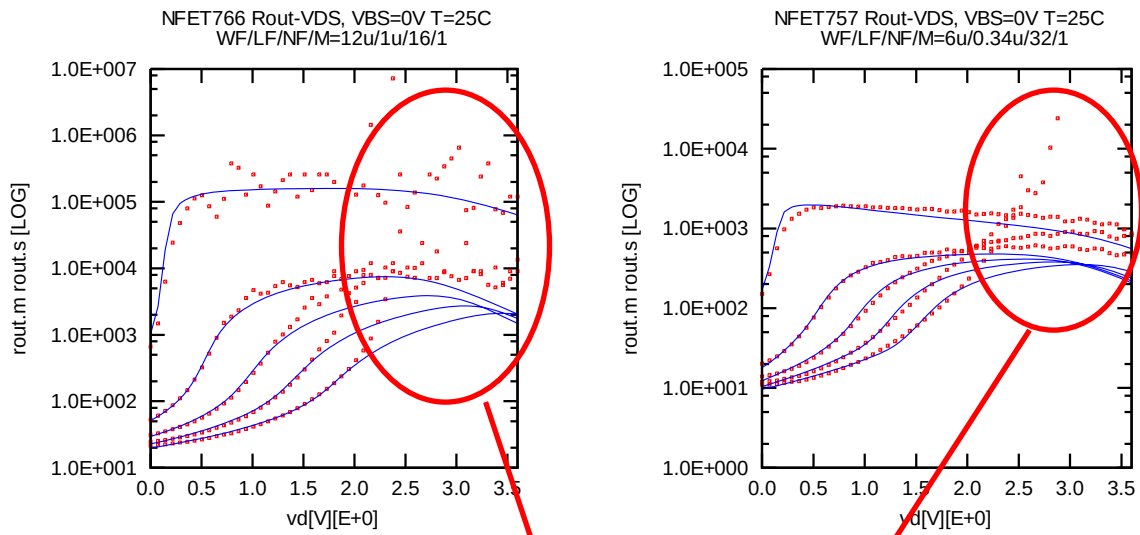


Figure 2-6 Id-Vd (25°C)



RF FET need large enough total gate width in order to get smooth and reliable S parameter measurement. However, too large gate width will cause self heating effect at high Vd and Vg bias, as the negative output resistance seen in the plots.

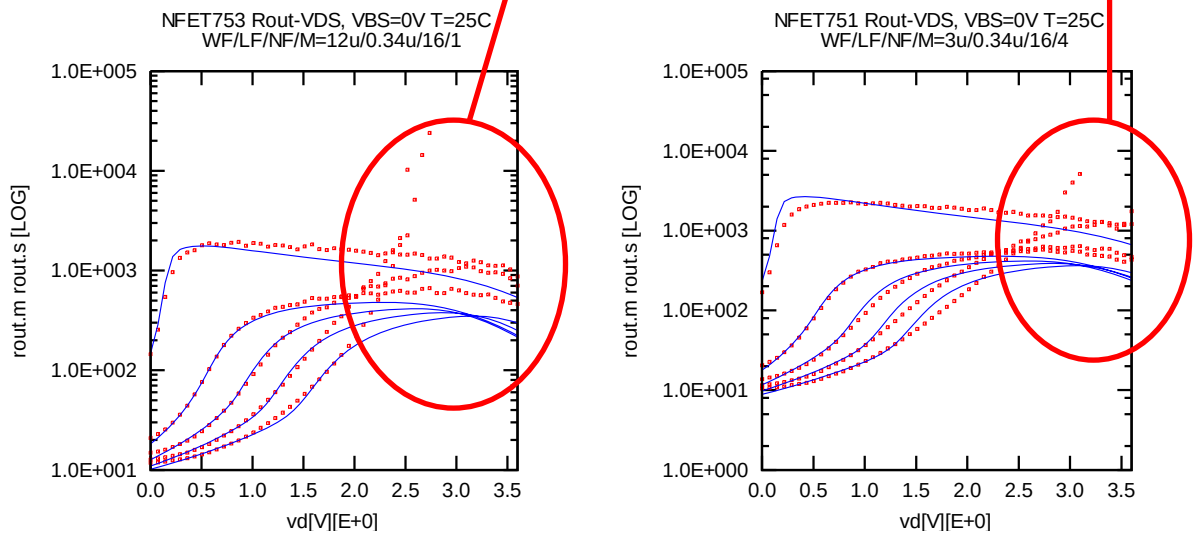


Figure 2-7 Rout-Vd (25°C)

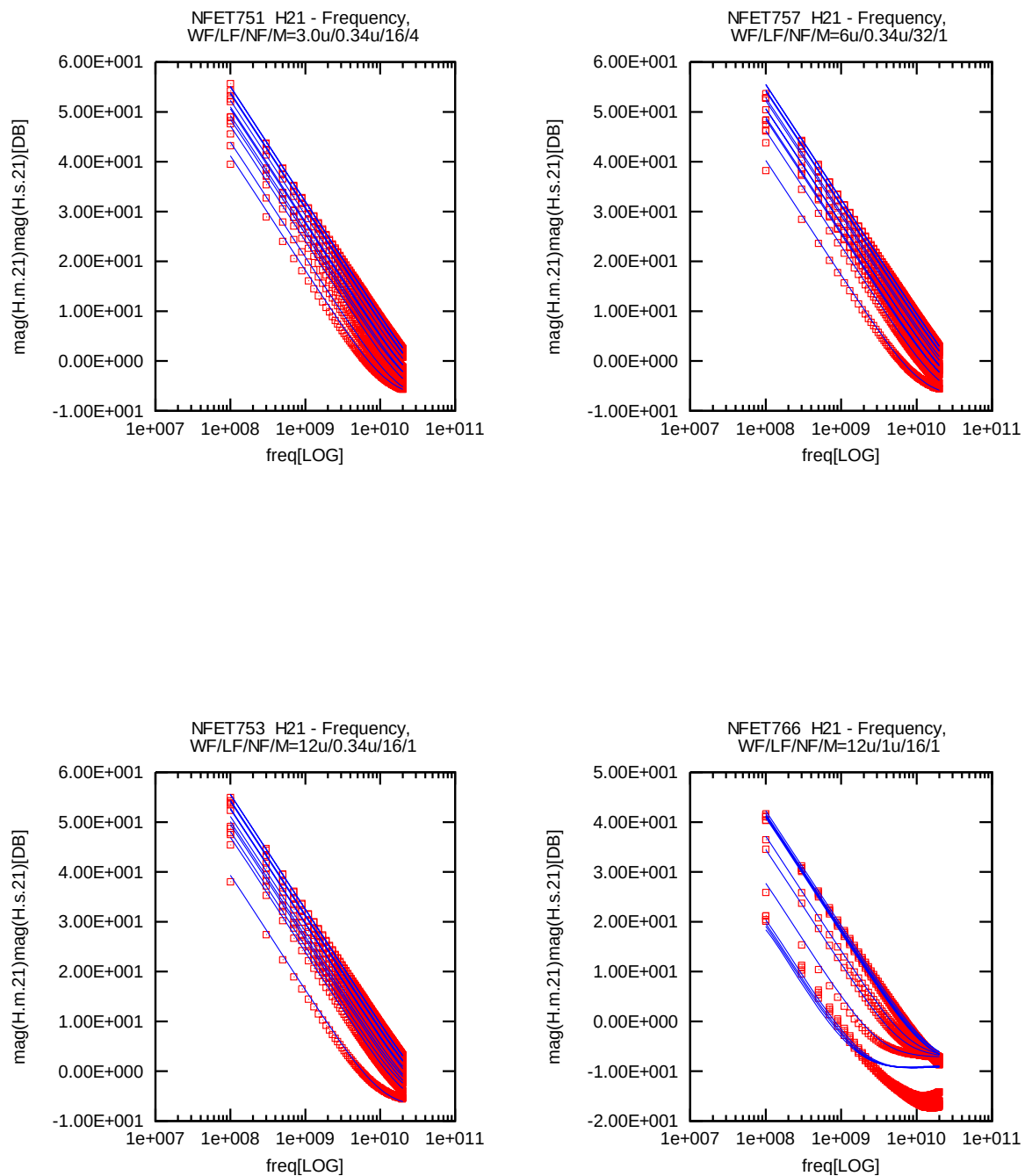


Figure 2-8 Mag(H21)-Freq (25°C)

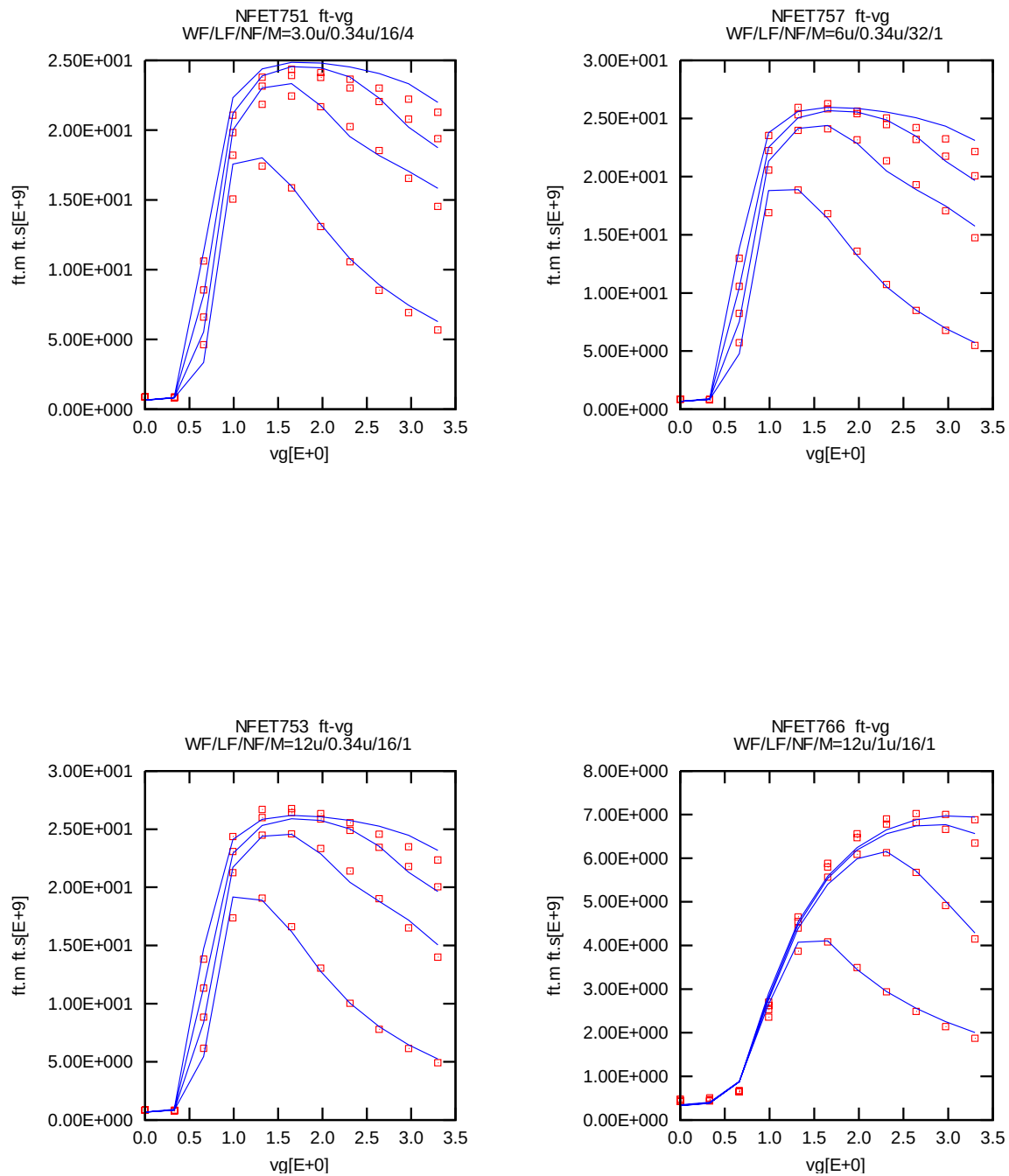


Figure 2-9 f_t - V_g (25°C)

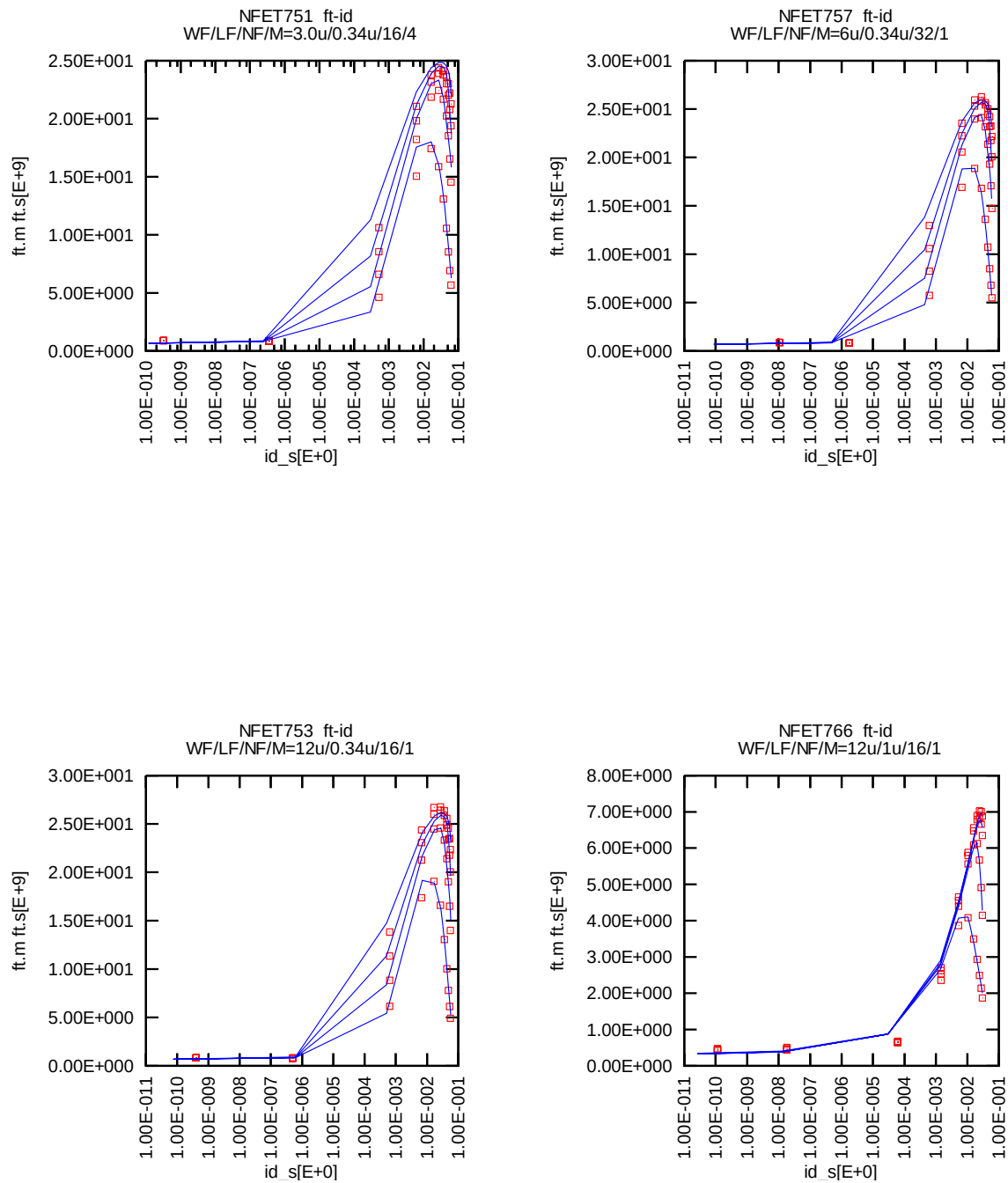
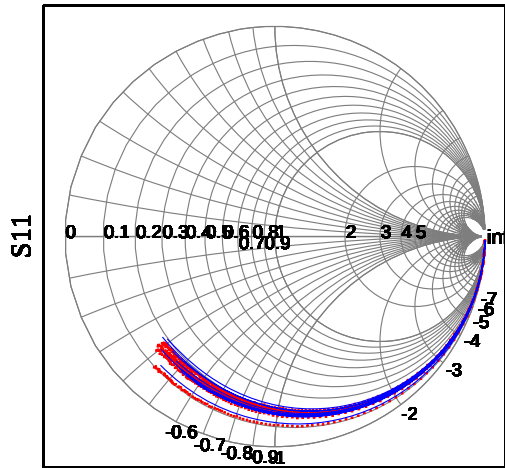
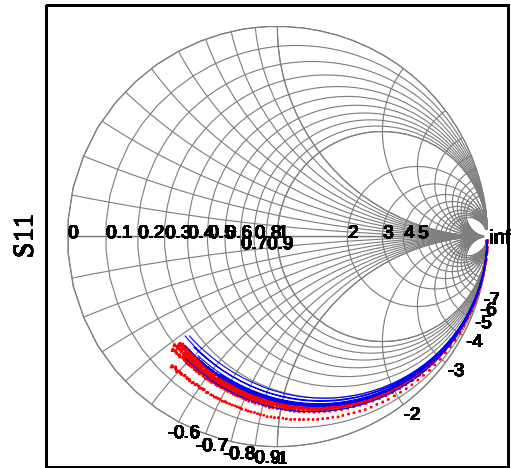


Figure 2-10 fT-Id (25°C)

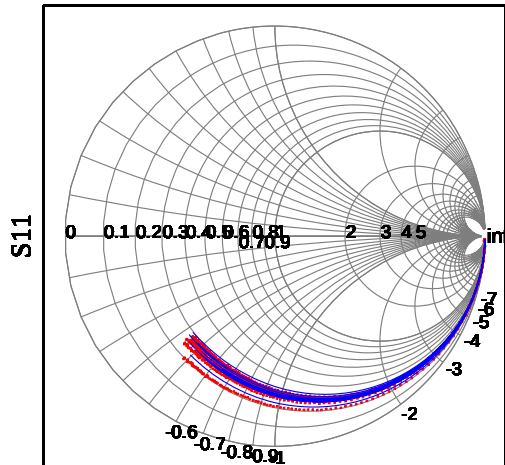
NFET751 S11 WF/LF/NF/M=3u/0.34u/16/4



NFET757 S11 WF/LF/NF/M=6u/0.34u/32/1



NFET753 S11 WF/LF/NF/M=12u/0.34u/16/1



NFET766 S11 WF/LF/NF/M=12u/1u/16/1

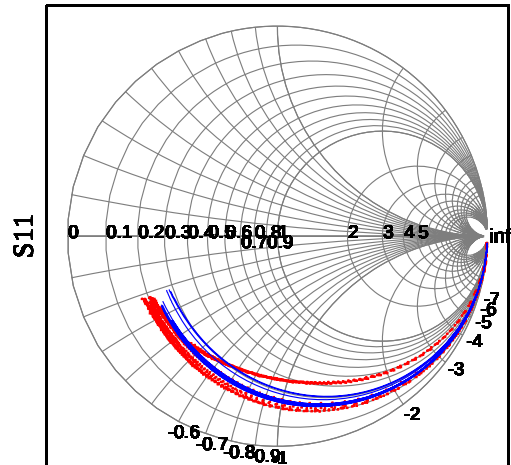


Figure 2-11 S11 (25°C)

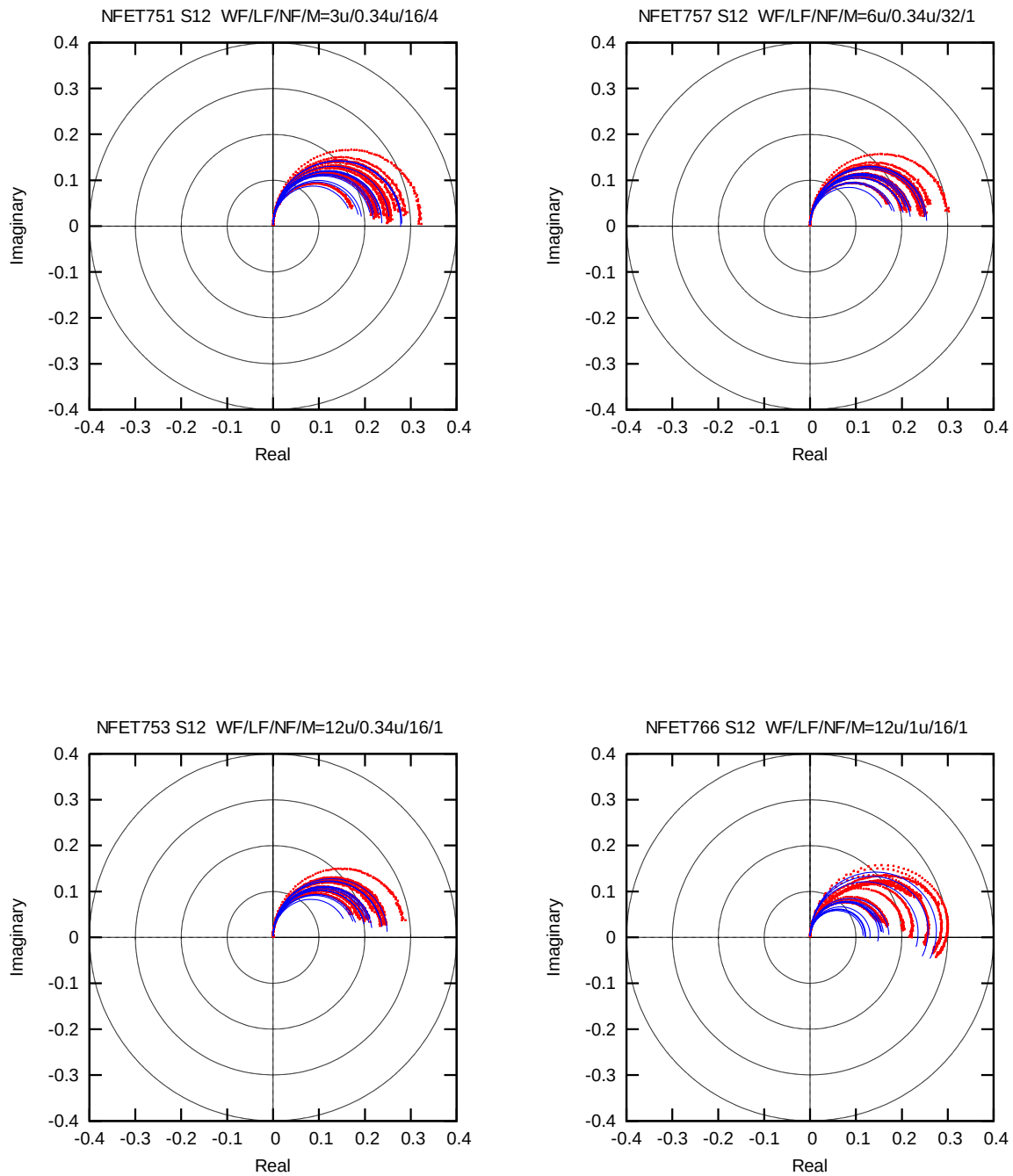


Figure 2-12 S12 (25°C)

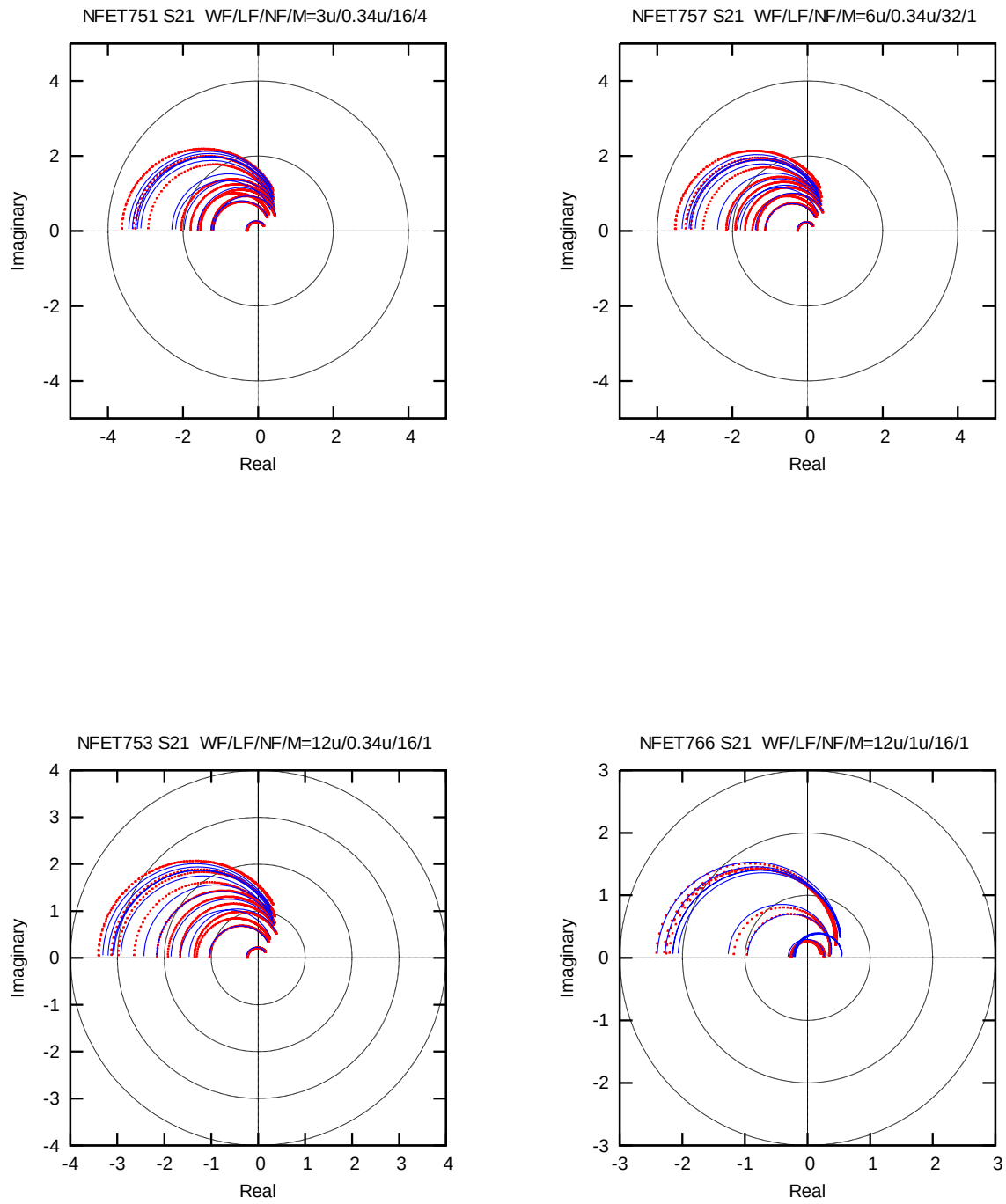
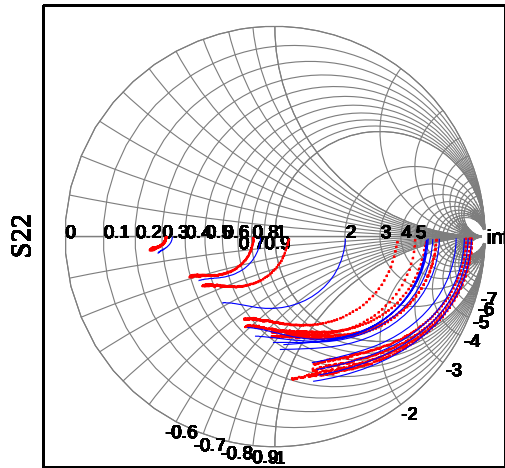
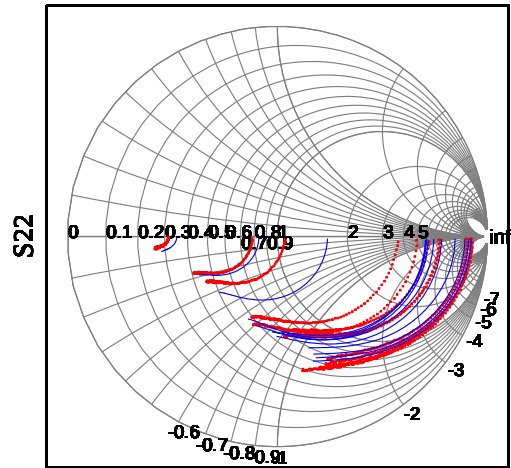


Figure 2-13 S21 (25°C)

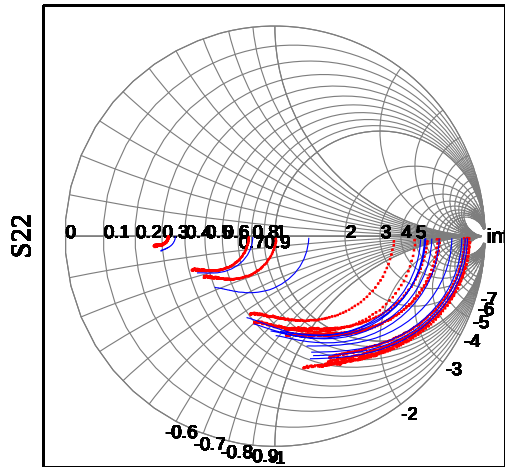
NFET751 S22 WF/LF/NF/M=3u/0.34u/16/4



NFET757 S22 WF/LF/NF/M=6u/0.34u/32/1



NFET753 S22 WF/LF/NF/M=12u/0.34u/16/1



NFET766 S22 WF/LF/NF/M=12u/1u/16/1

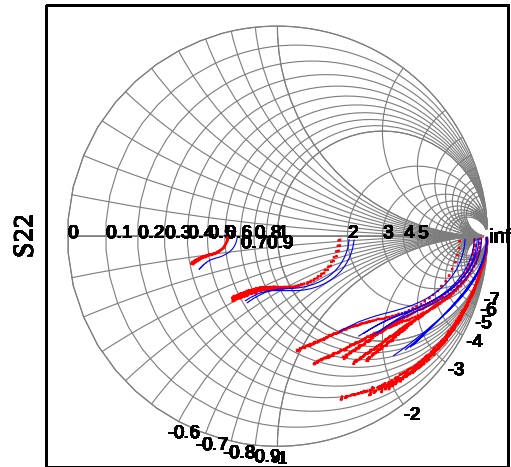


Figure 2-14 S22 (25°C)

2.2 PMOS Model

2.2.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing condition for the parameters are:

ft : Cut-off frequency extracted @ Ft_max:Vd=-3.3V, sweep Vg
vth : Linear Vth extracted @ Vd=-0.1V @ Id=-
70nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=-3.3V using fixed current Id=-
70nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ 3.3V
Idsat : Drain current extracted @ Vd=Vg=-3.3V
Idlin : Drain current extracted @ Vd=-0.1, Vg=-3.3V
Ioff : Drain current extracted @ Vg=0V, Vd=-3.3V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	10	10	5	2.5	10	10	5	2.5
	NF	16	16	8	16	16	16	8	16
SS	ft(GHz)	2.231	11.137	10.76	9.0308	NA	NA	NA	NA
	idsat(uA/um)	-8.50E+01	-2.28E+02	-2.35E+02	-2.32E+02	-8.35E+01	-2.46E+02	-2.43E+02	-2.37E+02
	vtlin(V)	-7.01E-01	-6.36E-01	-6.36E-01	-6.43E-01	-6.62E-01	-6.34E-01	-6.37E-01	-6.43E-01
	vtsat(V)	-6.89E-01	-5.93E-01	-5.92E-01	-5.99E-01	-6.52E-01	-5.73E-01	-5.76E-01	-5.82E-01
	body_effect(V)	-9.74E-01	-6.11E-01	-6.11E-01	-6.09E-01	-9.67E-01	-6.05E-01	-6.05E-01	-6.03E-01
	ioff(pA/um)	-7.82E-01	-8.35E-01	-1.71E+00	-3.17E+00	-7.97E-02	-1.98E-01	-2.62E-01	-3.92E-01
SNFP	ft(GHz)	2.6669	14.837	15.034	13.501	NA	NA	NA	NA
	idsat(uA/um)	-9.99E+01	-2.85E+02	-2.96E+02	-2.94E+02	-1.00E+02	-3.11E+02	-3.08E+02	-3.04E+02
	vtlin(V)	-6.16E-01	-5.20E-01	-5.19E-01	-5.25E-01	-5.74E-01	-5.13E-01	-5.15E-01	-5.18E-01
	vtsat(V)	-6.05E-01	-4.76E-01	-4.76E-01	-4.81E-01	-5.64E-01	-4.51E-01	-4.53E-01	-4.57E-01
	body_effect(V)	-9.56E-01	-5.71E-01	-5.71E-01	-5.69E-01	-9.48E-01	-5.66E-01	-5.65E-01	-5.63E-01
	ioff(pA/um)	-7.93E-01	-1.93E+00	-2.81E+00	-4.12E+00	-1.30E-01	-3.01E+00	-2.94E+00	-2.83E+00
TT	ft(GHz)	2.4981	13.324	13.332	11.697	NA	NA	NA	NA
	idsat(uA/um)	-9.50E+01	-2.66E+02	-2.76E+02	-2.73E+02	-9.45E+01	-2.89E+02	-2.86E+02	-2.82E+02
	vtlin(V)	-6.44E-01	-5.57E-01	-5.58E-01	-5.64E-01	-6.03E-01	-5.53E-01	-5.55E-01	-5.59E-01
	vtsat(V)	-6.33E-01	-5.15E-01	-5.15E-01	-5.21E-01	-5.93E-01	-4.92E-01	-4.94E-01	-4.99E-01
	body_effect(V)	-9.56E-01	-5.85E-01	-5.85E-01	-5.83E-01	-9.49E-01	-5.79E-01	-5.79E-01	-5.77E-01
	ioff(pA/um)	-7.87E-01	-1.19E+00	-2.06E+00	-3.46E+00	-9.99E-02	-1.06E+00	-1.08E+00	-1.12E+00
FNFP	ft(GHz)	2.3212	11.834	11.65	9.936	NA	NA	NA	NA
	idsat(uA/um)	-9.01E+01	-2.47E+02	-2.56E+02	-2.51E+02	-8.90E+01	-2.67E+02	-2.64E+02	-2.59E+02
	vtlin(V)	-6.72E-01	-5.95E-01	-5.95E-01	-6.03E-01	-6.32E-01	-5.92E-01	-5.95E-01	-6.00E-01
	vtsat(V)	-6.60E-01	-5.53E-01	-5.53E-01	-5.61E-01	-6.22E-01	-5.33E-01	-5.36E-01	-5.42E-01
	body_effect(V)	-9.56E-01	-5.98E-01	-5.98E-01	-5.96E-01	-9.49E-01	-5.92E-01	-5.92E-01	-5.90E-01
	ioff(pA/um)	-7.83E-01	-9.24E-01	-1.80E+00	-3.24E+00	-8.61E-02	-4.04E-01	-4.56E-01	-5.62E-01
FF	ft(GHz)	2.7303	15.373	15.804	14.381	NA	NA	NA	NA
	idsat(uA/um)	-1.05E+02	-3.04E+02	-3.17E+02	-3.12E+02	-1.05E+02	-3.32E+02	-3.30E+02	-3.26E+02
	vtlin(V)	-5.87E-01	-4.80E-01	-4.80E-01	-4.85E-01	-5.44E-01	-4.71E-01	-4.73E-01	-4.75E-01
	vtsat(V)	-5.76E-01	-4.38E-01	-4.37E-01	-4.43E-01	-5.34E-01	-4.10E-01	-4.12E-01	-4.15E-01
	body_effect(V)	-9.37E-01	-5.59E-01	-5.59E-01	-5.57E-01	-9.30E-01	-5.53E-01	-5.53E-01	-5.51E-01
	ioff(pA/um)	-8.07E-01	-3.84E+00	-4.74E+00	-5.82E+00	-1.89E-01	-8.22E+00	-7.97E+00	-7.55E+00

Table 2-5 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	10	10	5	2.5	10	10	5	2.5
	NF	16	16	8	16	16	16	8	16
SS	ft(GHz)	1.9088	9.4762	9.2065	7.7689	NA	NA	NA	NA
	idsat(uA/um)	-7.38E+01	-1.99E+02	-2.06E+02	-2.04E+02	-7.16E+01	-2.15E+02	-2.12E+02	-2.06E+02
	vtlin(V)	-5.78E-01	-5.15E-01	-5.15E-01	-5.23E-01	-5.41E-01	-5.14E-01	-5.18E-01	-5.23E-01
	vtsat(V)	-5.62E-01	-4.67E-01	-4.66E-01	-4.74E-01	-5.28E-01	-4.50E-01	-4.53E-01	-4.60E-01
	body_effect(V)	-1.02E+00	-6.53E-01	-6.53E-01	-6.52E-01	-1.01E+00	-6.45E-01	-6.45E-01	-6.43E-01
	ioff(pA/um)	-2.42E+00	-3.13E+01	-3.24E+01	-2.98E+01	-4.88E+00	-5.23E+01	-4.93E+01	-4.40E+01
SNFP	ft(GHz)	2.274	12.495	12.714	11.464	NA	NA	NA	NA
	idsat(uA/um)	-8.63E+01	-2.47E+02	-2.58E+02	-2.56E+02	-8.53E+01	-2.69E+02	-2.67E+02	-2.62E+02
	vtlin(V)	-4.91E-01	-3.96E-01	-3.96E-01	-4.02E-01	-4.51E-01	-3.91E-01	-3.93E-01	-3.97E-01
	vtsat(V)	-4.76E-01	-3.49E-01	-3.48E-01	-3.54E-01	-4.39E-01	-3.25E-01	-3.28E-01	-3.32E-01
	body_effect(V)	-1.00E+00	-6.15E-01	-6.15E-01	-6.13E-01	-9.91E-01	-6.07E-01	-6.06E-01	-6.04E-01
	ioff(pA/um)	-9.45E+00	-3.11E+02	-3.17E+02	-2.83E+02	-2.33E+01	-5.69E+02	-5.46E+02	-5.06E+02
TT	ft(GHz)	2.134	11.257	11.314	9.9716	NA	NA	NA	NA
	idsat(uA/um)	-8.23E+01	-2.31E+02	-2.40E+02	-2.38E+02	-8.08E+01	-2.51E+02	-2.48E+02	-2.43E+02
	vtlin(V)	-5.20E-01	-4.35E-01	-4.35E-01	-4.42E-01	-4.81E-01	-4.31E-01	-4.34E-01	-4.38E-01
	vtsat(V)	-5.05E-01	-3.89E-01	-3.88E-01	-3.95E-01	-4.68E-01	-3.67E-01	-3.70E-01	-3.75E-01
	body_effect(V)	-1.00E+00	-6.28E-01	-6.28E-01	-6.26E-01	-9.91E-01	-6.20E-01	-6.20E-01	-6.18E-01
	ioff(pA/um)	-5.67E+00	-1.42E+02	-1.44E+02	-1.28E+02	-1.34E+01	-2.51E+02	-2.39E+02	-2.18E+02
FNFP	ft(GHz)	1.9872	10.033	9.9246	8.507	NA	NA	NA	NA
	idsat(uA/um)	-7.82E+01	-2.15E+02	-2.23E+02	-2.20E+02	-7.62E+01	-2.32E+02	-2.30E+02	-2.24E+02
	vtlin(V)	-5.48E-01	-4.73E-01	-4.74E-01	-4.82E-01	-5.11E-01	-4.72E-01	-4.75E-01	-4.80E-01
	vtsat(V)	-5.33E-01	-4.28E-01	-4.27E-01	-4.35E-01	-4.98E-01	-4.10E-01	-4.13E-01	-4.18E-01
	body_effect(V)	-1.00E+00	-6.41E-01	-6.41E-01	-6.39E-01	-9.92E-01	-6.33E-01	-6.33E-01	-6.31E-01
	ioff(pA/um)	-3.55E+00	-6.54E+01	-6.65E+01	-5.89E+01	-7.82E+00	-1.10E+02	-1.04E+02	-9.38E+01
FF	ft(GHz)	2.332	12.911	13.32	12.165	NA	NA	NA	NA
	idsat(uA/um)	-9.06E+01	-2.63E+02	-2.74E+02	-2.70E+02	-8.99E+01	-2.87E+02	-2.85E+02	-2.80E+02
	vtlin(V)	-4.62E-01	-3.56E-01	-3.55E-01	-3.62E-01	-4.21E-01	-3.49E-01	-3.50E-01	-3.53E-01
	vtsat(V)	-4.47E-01	-3.10E-01	-3.09E-01	-3.15E-01	-4.09E-01	-2.84E-01	-2.86E-01	-2.89E-01
	body_effect(V)	-9.83E-01	-6.03E-01	-6.03E-01	-6.00E-01	-9.73E-01	-5.95E-01	-5.95E-01	-5.93E-01
	ioff(pA/um)	-1.56E+01	-6.56E+02	-6.68E+02	-5.96E+02	-3.97E+01	-1.23E+03	-1.19E+03	-1.12E+03

Table 2-6 RF & Logic Corner Table at 125 °C

Temp=-55C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	10	10	5	2.5	10	10	5	2.5
	NF	16	16	8	16	16	16	8	16
SS	ft(GHz)	2.639	13.106	12.572	10.492	NA	NA	NA	NA
	idsat(uA/um)	-9.98E+01	-2.63E+02	-2.69E+02	-2.65E+02	-9.87E+01	-2.82E+02	-2.79E+02	-2.73E+02
	vtlin(V)	-8.04E-01	-7.36E-01	-7.37E-01	-7.44E-01	-7.64E-01	-7.35E-01	-7.38E-01	-7.43E-01
	vtsat(V)	-7.94E-01	-6.95E-01	-6.95E-01	-7.02E-01	-7.55E-01	-6.76E-01	-6.79E-01	-6.84E-01
	body_effect(V)	-9.40E-01	-5.78E-01	-5.78E-01	-5.77E-01	-9.34E-01	-5.74E-01	-5.74E-01	-5.72E-01
	ioff(pA/um)	-7.80E-01	-7.80E-01	-1.65E+00	-3.12E+00	-7.26E-02	-7.26E-02	-1.47E-01	-2.93E-01
SNFP	ft(GHz)	3.1685	17.663	17.806	15.933	NA	NA	NA	NA
	idsat(uA/um)	-1.18E+02	-3.32E+02	-3.44E+02	-3.41E+02	-1.19E+02	-3.60E+02	-3.58E+02	-3.54E+02
	vtlin(V)	-7.20E-01	-6.22E-01	-6.22E-01	-6.27E-01	-6.78E-01	-6.15E-01	-6.17E-01	-6.20E-01
	vtsat(V)	-7.10E-01	-5.81E-01	-5.80E-01	-5.85E-01	-6.69E-01	-5.54E-01	-5.56E-01	-5.60E-01
	body_effect(V)	-9.21E-01	-5.39E-01	-5.38E-01	-5.37E-01	-9.15E-01	-5.35E-01	-5.34E-01	-5.32E-01
	ioff(pA/um)	-7.80E-01	-7.81E-01	-1.65E+00	-3.12E+00	-7.26E-02	-7.46E-02	-1.48E-01	-2.94E-01
TT	ft(GHz)	2.9626	15.819	15.742	13.752	NA	NA	NA	NA
	idsat(uA/um)	-1.12E+02	-3.09E+02	-3.20E+02	-3.15E+02	-1.12E+02	-3.34E+02	-3.32E+02	-3.27E+02
	vtlin(V)	-7.48E-01	-6.59E-01	-6.60E-01	-6.66E-01	-7.06E-01	-6.54E-01	-6.56E-01	-6.60E-01
	vtsat(V)	-7.38E-01	-6.19E-01	-6.19E-01	-6.25E-01	-6.97E-01	-5.95E-01	-5.97E-01	-6.02E-01
	body_effect(V)	-9.22E-01	-5.52E-01	-5.52E-01	-5.51E-01	-9.16E-01	-5.48E-01	-5.48E-01	-5.46E-01
	ioff(pA/um)	-7.80E-01	-7.80E-01	-1.65E+00	-3.12E+00	-7.30E-02	-7.34E-02	-1.46E-01	-2.94E-01
FNFP	ft(GHz)	2.747	14.005	13.704	11.631	NA	NA	NA	NA
	idsat(uA/um)	-1.06E+02	-2.86E+02	-2.95E+02	-2.89E+02	-1.05E+02	-3.08E+02	-3.05E+02	-3.00E+02
	vtlin(V)	-7.75E-01	-6.96E-01	-6.97E-01	-7.04E-01	-7.35E-01	-6.94E-01	-6.96E-01	-7.01E-01
	vtsat(V)	-7.65E-01	-6.57E-01	-6.57E-01	-6.64E-01	-7.25E-01	-6.36E-01	-6.39E-01	-6.44E-01
	body_effect(V)	-9.22E-01	-5.66E-01	-5.66E-01	-5.64E-01	-9.16E-01	-5.61E-01	-5.61E-01	-5.59E-01
	ioff(pA/um)	-7.80E-01	-7.80E-01	-1.65E+00	-3.12E+00	-7.30E-02	-7.30E-02	-1.47E-01	-2.93E-01
FF	ft(GHz)	3.2407	18.378	18.818	17.077	NA	NA	NA	NA
	idsat(uA/um)	-1.24E+02	-3.55E+02	-3.69E+02	-3.64E+02	-1.25E+02	-3.86E+02	-3.84E+02	-3.81E+02
	vtlin(V)	-6.91E-01	-5.82E-01	-5.82E-01	-5.87E-01	-6.48E-01	-5.74E-01	-5.75E-01	-5.77E-01
	vtsat(V)	-6.82E-01	-5.42E-01	-5.42E-01	-5.47E-01	-6.39E-01	-5.14E-01	-5.16E-01	-5.19E-01
	body_effect(V)	-9.03E-01	-5.27E-01	-5.26E-01	-5.24E-01	-8.96E-01	-5.22E-01	-5.22E-01	-5.20E-01
	ioff(pA/um)	-7.80E-01	-7.81E-01	-1.65E+00	-3.12E+00	-7.26E-02	-7.86E-02	-1.52E-01	-2.98E-01

Table 2-7 RF & Logic Corner Table at -55 °C

2.2.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device					Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
PFET801	0.3	2.5	16	5	-0.571	-0.548	-0.493	-0.495	-306.62	-306.57	-9.55	-4.02	14.09	13.994
PFET802	0.3	5	16	4	-0.569	-0.544	-0.491	-0.491	-302.97	-299.26	-5.56	-2.561	15.341	15.572
PFET803	0.3	10	16	2	-0.567	-0.541	-0.489	-0.489	-294.92	-295.79	-5.3	-1.834	15.807	15.769
PFET804	0.3	15	16	1	-0.564	-0.541	-0.487	-0.488	-289.69	-294.66	-5.89	-1.59	15.923	15.769
PFET805	0.3	10	4	8	-0.567	-0.537	-0.489	-0.483	-316.66	-317.68	-4.7	-2.09	16.451	16.063
PFET806	0.3	10	8	4	-0.568	-0.539	-0.49	-0.486	-303.63	-306.68	-4.36	-1.937	16.076	16.055
PFET807	0.3	10	32	1	-0.569	-0.544	-0.491	-0.491	-290.24	-284.5	-4.2	-1.756	15.71	15.262
PFET810	0.3	5	8	2	-0.567	-0.541	-0.488	-0.488	-316.36	-310.28	-11.04	-2.708	15.886	15.915
PFET812	0.3	5	32	1	-0.568	-0.546	-0.489	-0.493	-303.03	-289.6	-6.58	-2.464	15.6	15.274
PFET813	0.34	10	16	1	-0.587	-0.562	-0.535	-0.519	-264.17	-262.03	-5.14	-1.185	13.57	13.119
PFET814	0.5	10	16	2	-0.633	-0.605	-0.599	-0.582	-178.62	-179.74	-2.33	-0.831	7.62	7.28
PFET815	0.7	10	16	2	-0.647	-0.629	-0.62	-0.614	-130.26	-130.95	-2.31	-0.796	4.564	4.356
PFET816	1	10	16	3	-0.655	-0.646	-0.631	-0.634	-92.53	-94.231	-1.51	-0.787	2.582	2.478
PFET820	0.34	10	4	2	-0.588	-0.557	-0.536	-0.514	-282.58	-281.76	-8.55	-1.366	13.986	13.468
PFET833	0.34	2.5	16	5	-0.59	-0.569	-0.538	-0.525	-266.43	-270.87	-3.77	-3.466	11.993	11.567
PFET834	0.34	5	8	2	-0.591	-0.562	-0.54	-0.519	-275.56	-274.45	-8.18	-2.056	13.263	13.225
PFET835	0.34	15	16	1	-0.587	-0.561	-0.535	-0.519	-255.78	-260.5	-3.17	-0.932	13.456	13.084

Table 2-8 Summary Table for RF PMOS

Measure: Measurement result

Model: Simulation result

2.2.3 Simulation results for WF, LF trends

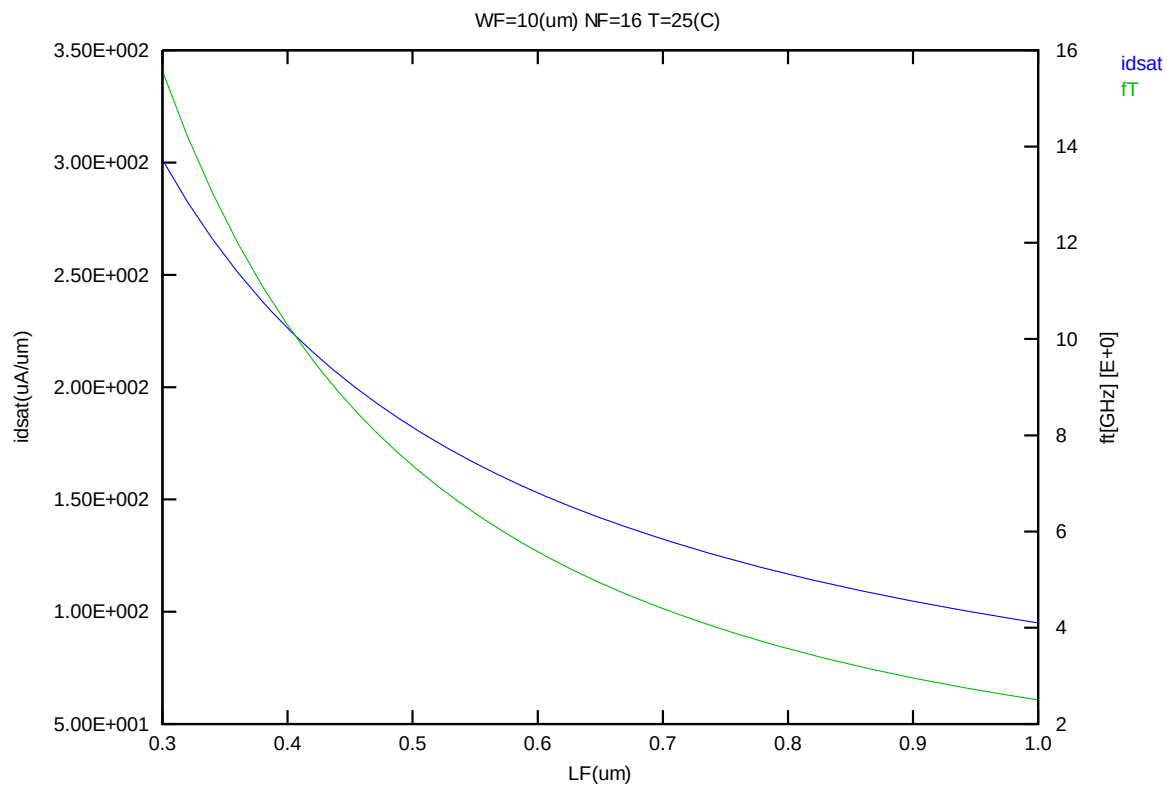


Figure 2-15 fT and idsat vs. channel length

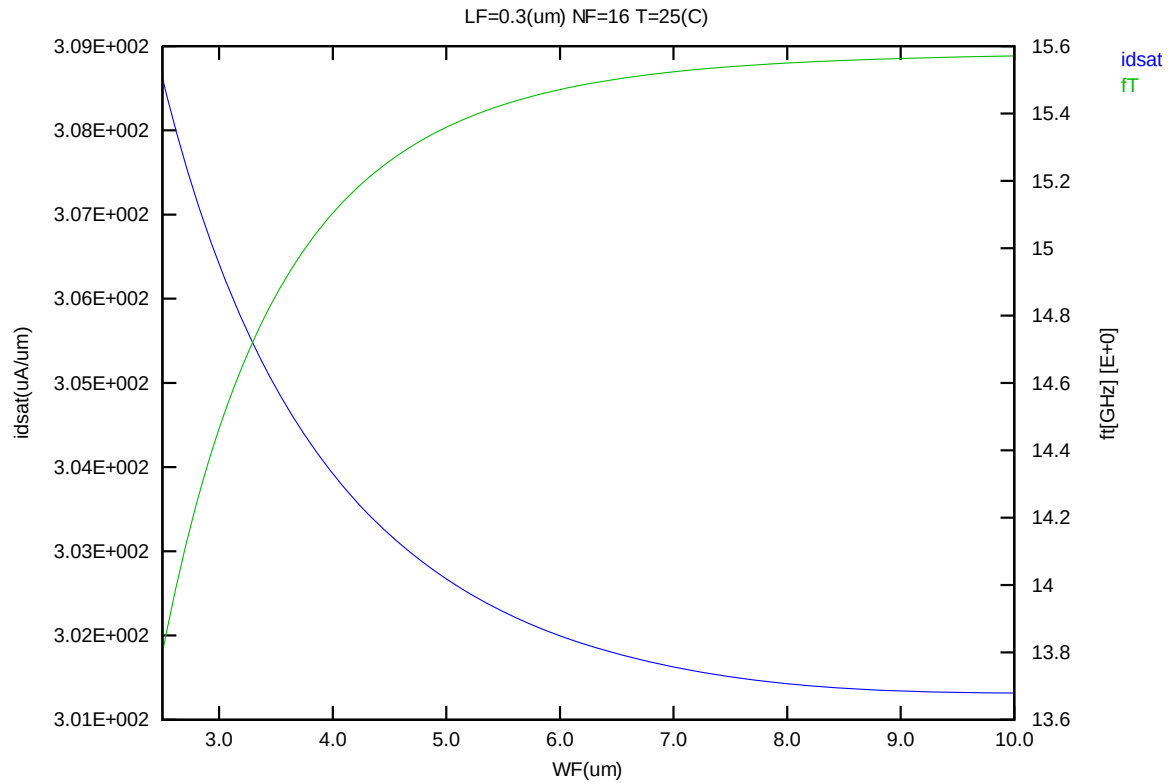


Figure 2-16 fT and idsat vs. channel width

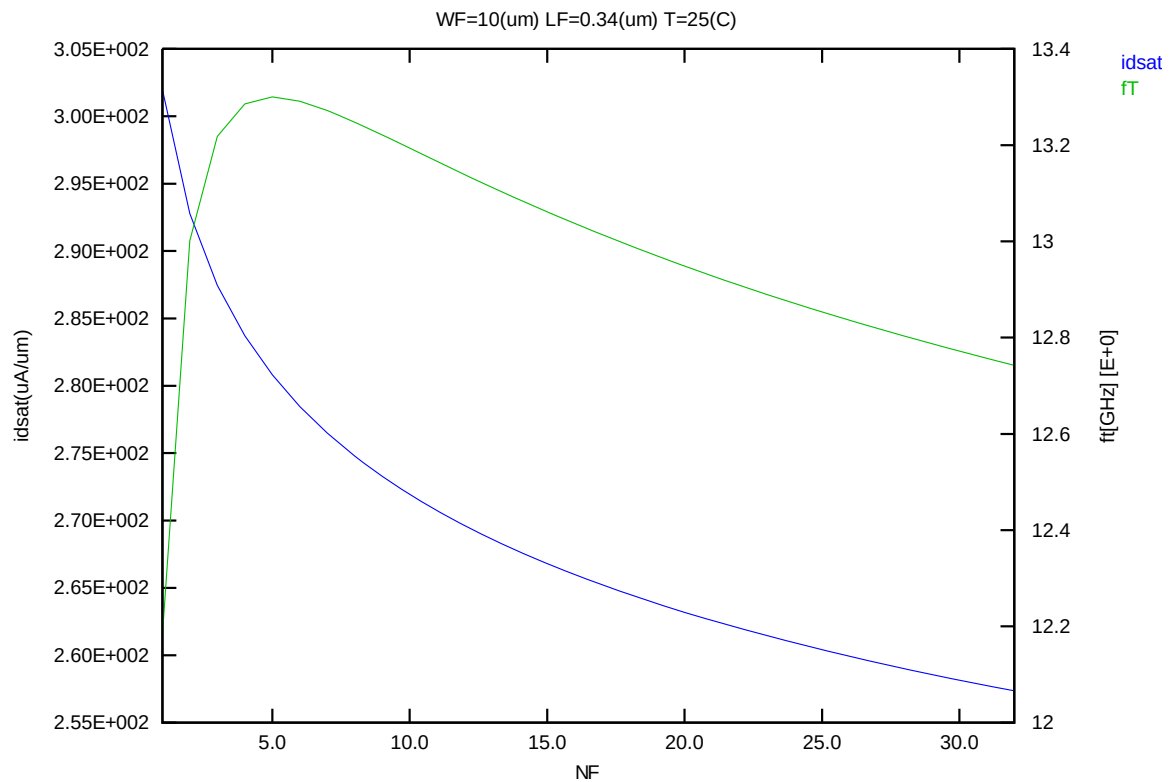


Figure 2-17 fT and idsat vs. finger number

2.2.4 DC & RF Fitting curves for individual devices

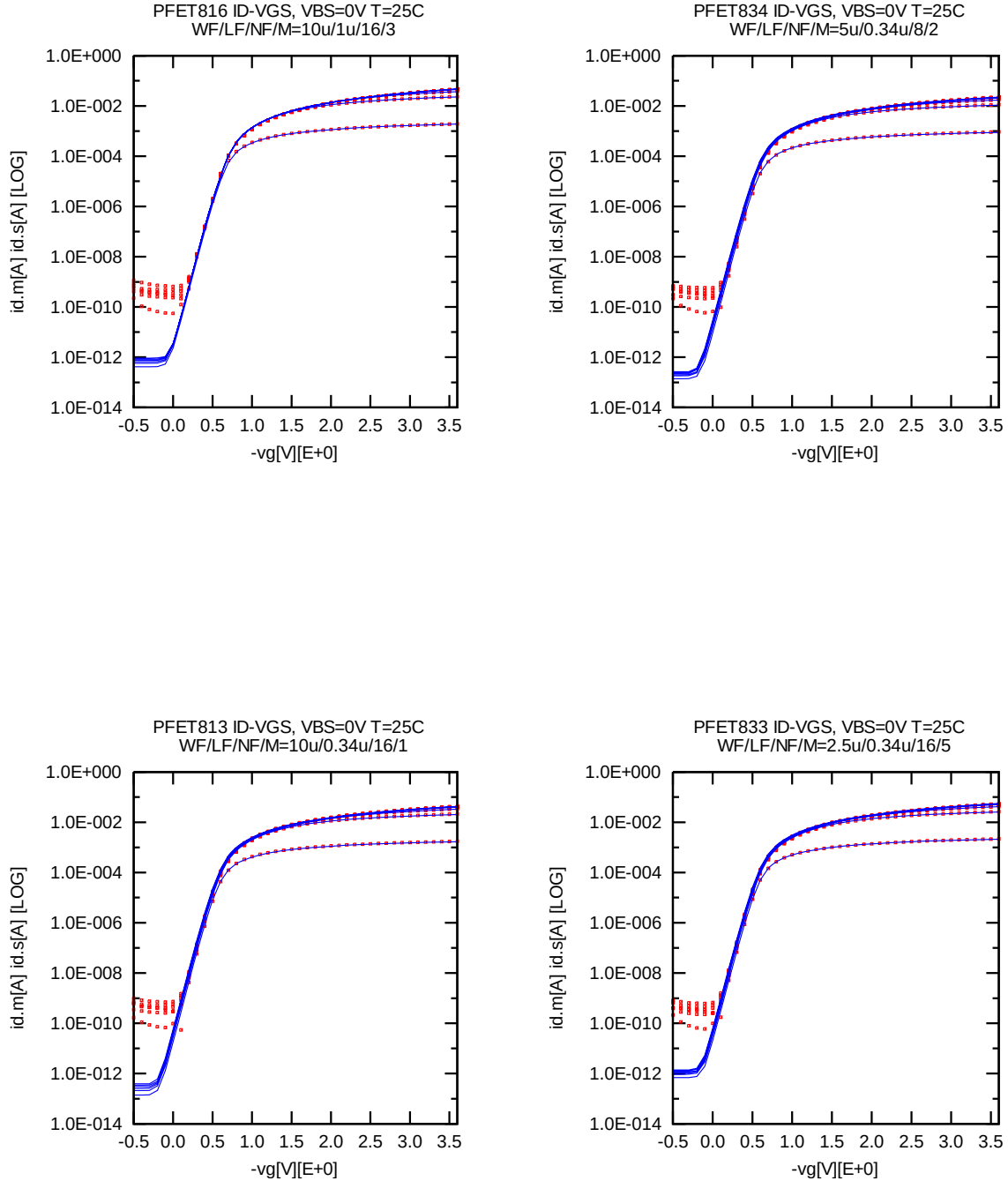


Figure 2-18 Id-Vg (25°C)

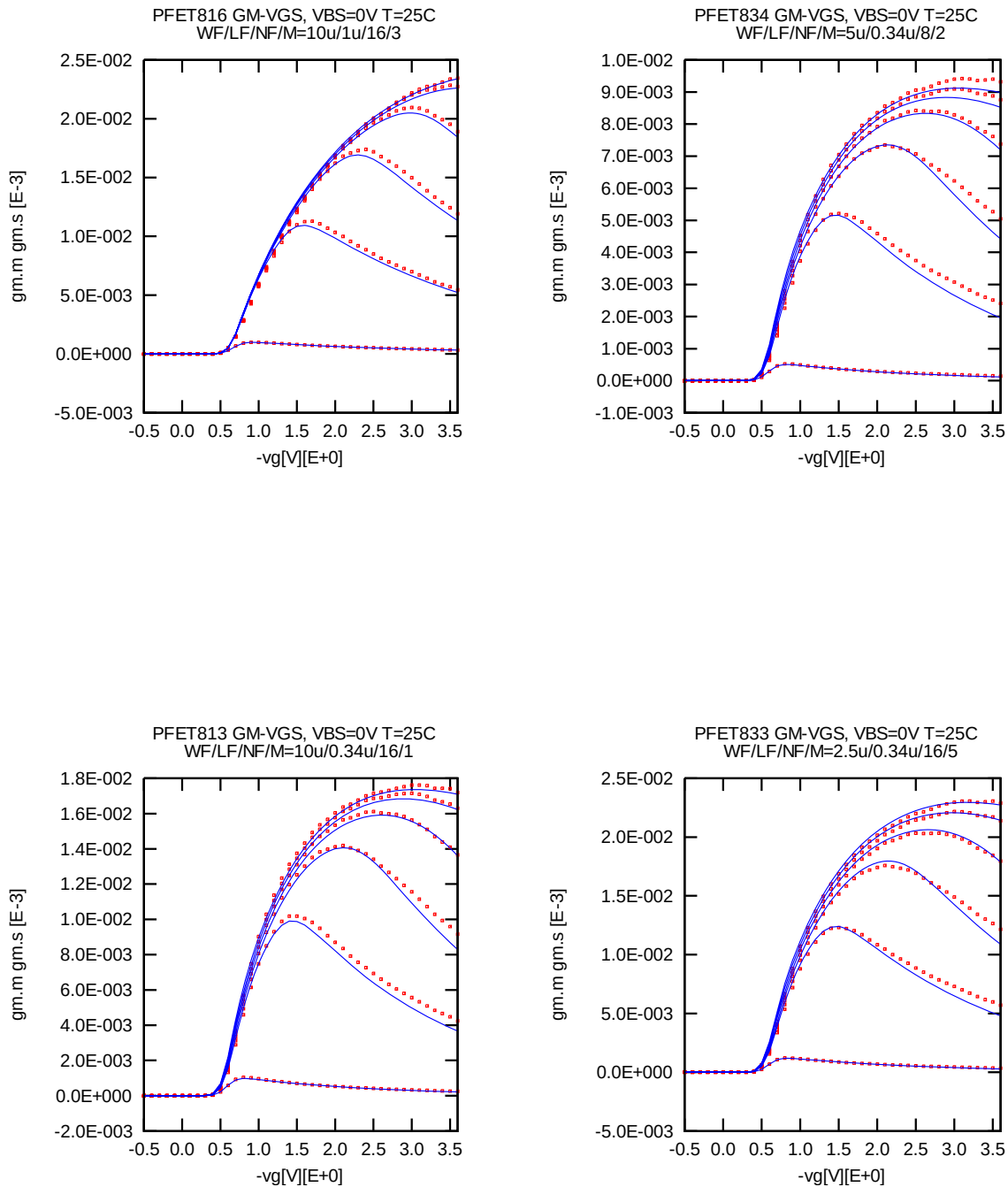


Figure 2-19 Gm-Vg (25°C)

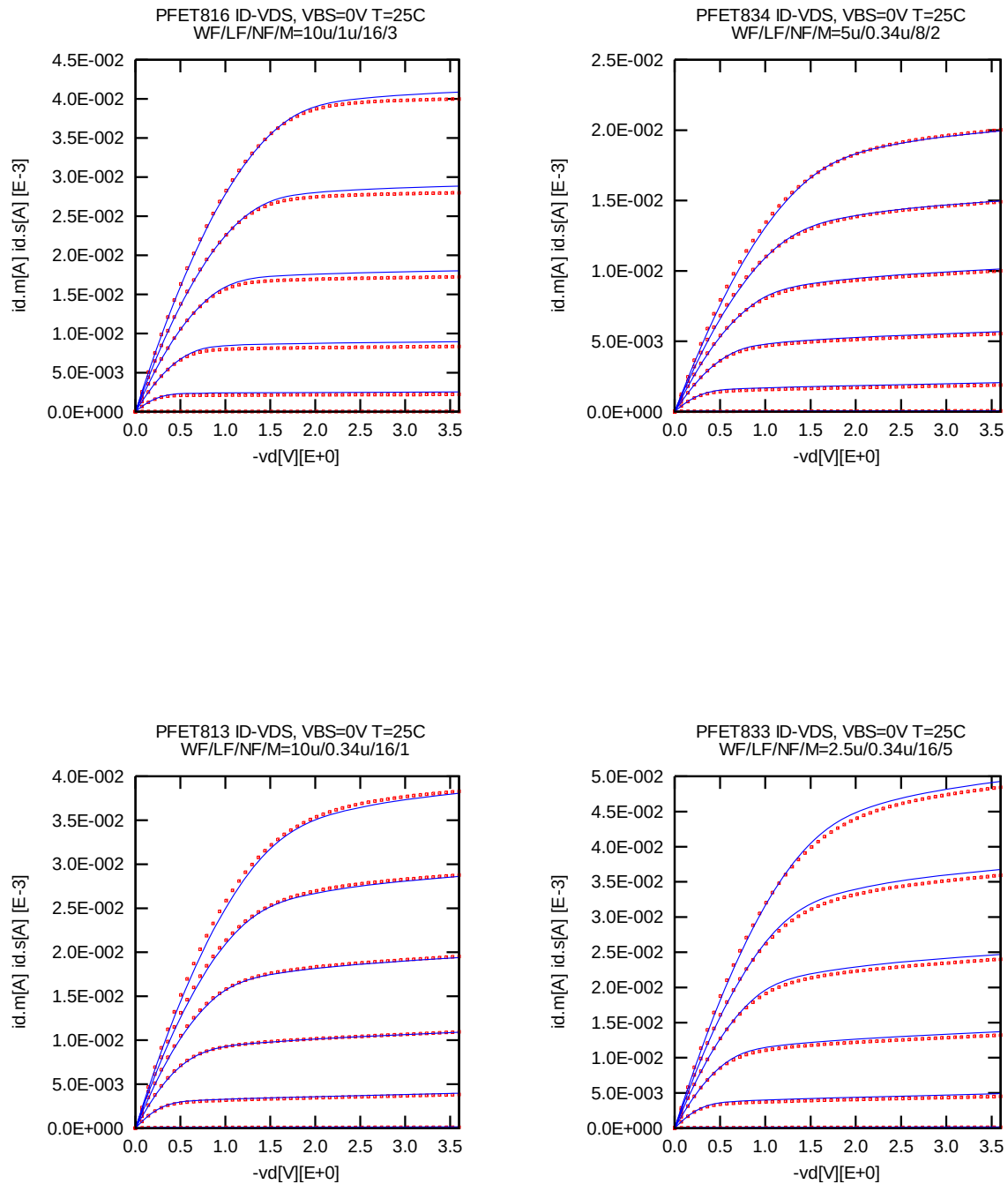


Figure 2-20 Id-Vd (25°C)

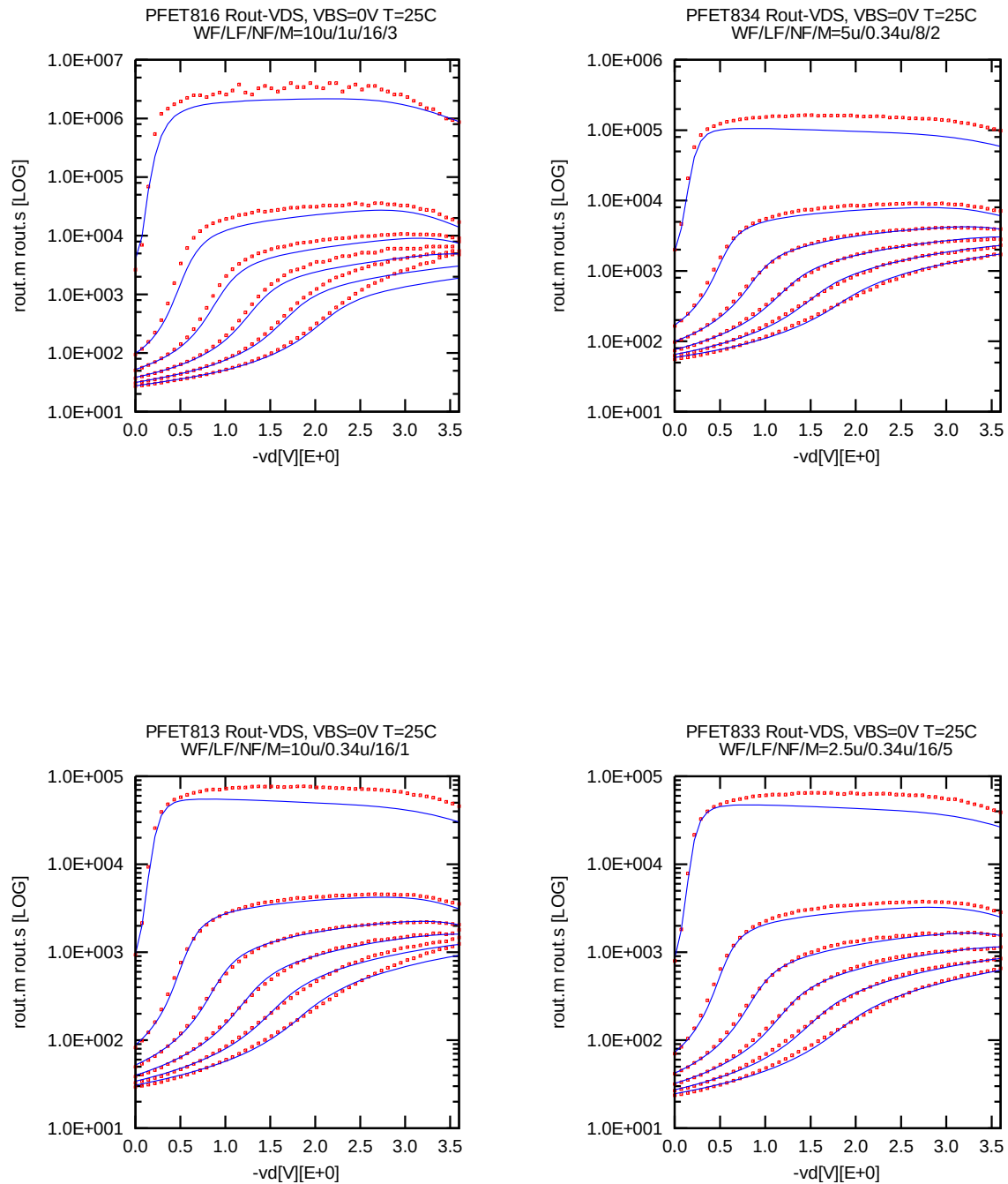


Figure 2-21 Rout-Vd (25°C)

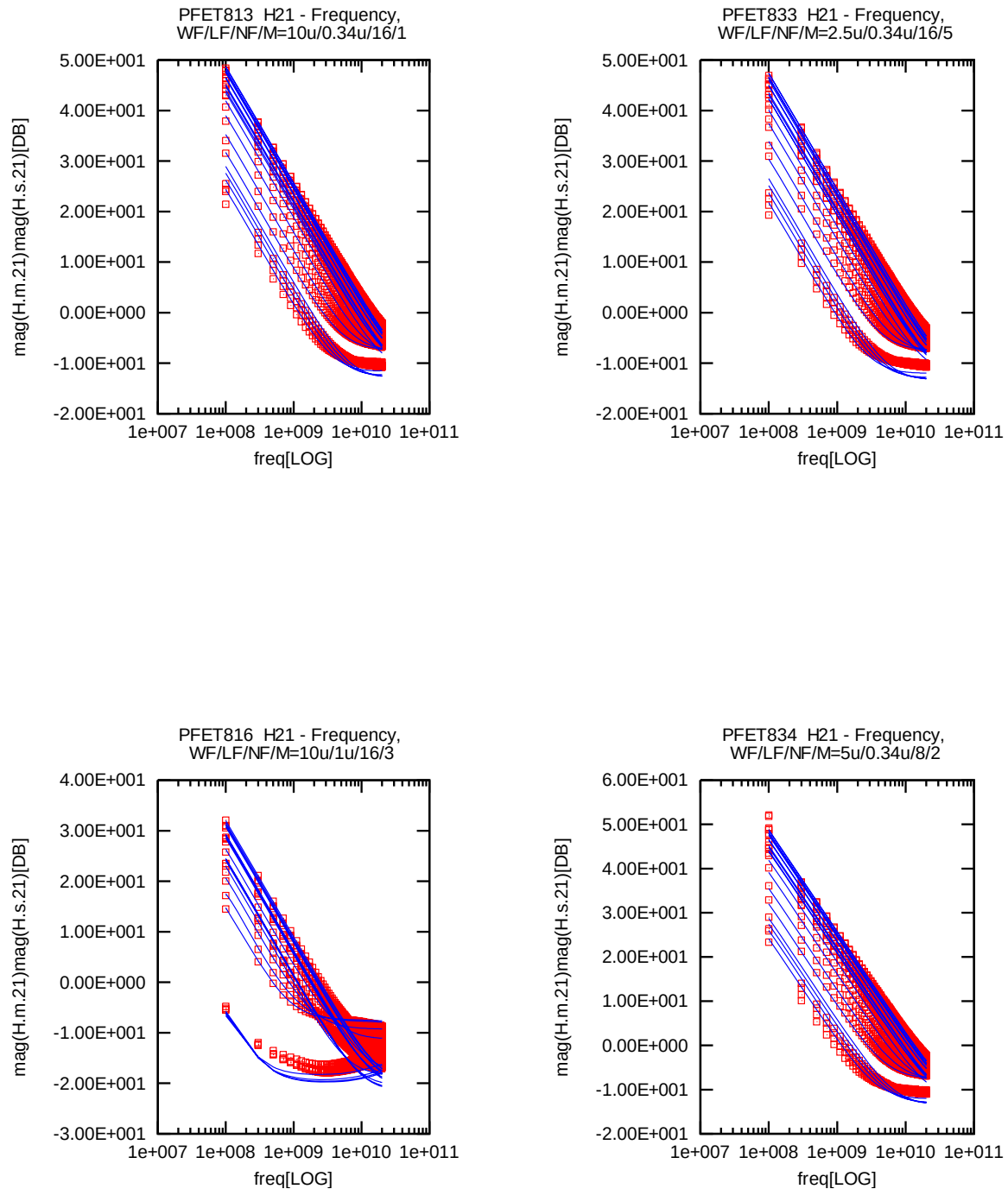


Figure 2-22 Mag(H21)-Freq (25°C)

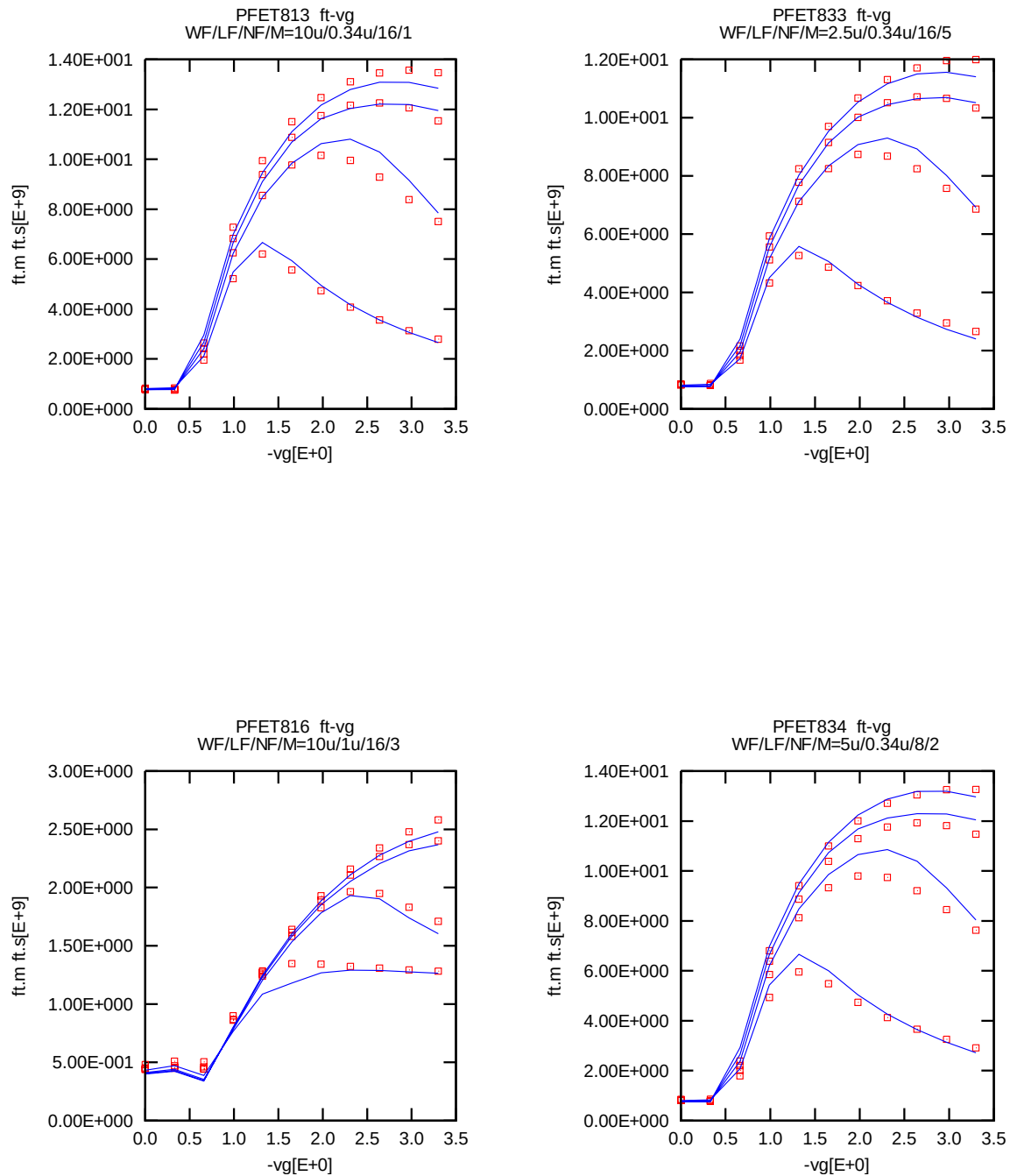


Figure 2-23 f_T -Vg ($25^\circ C$)

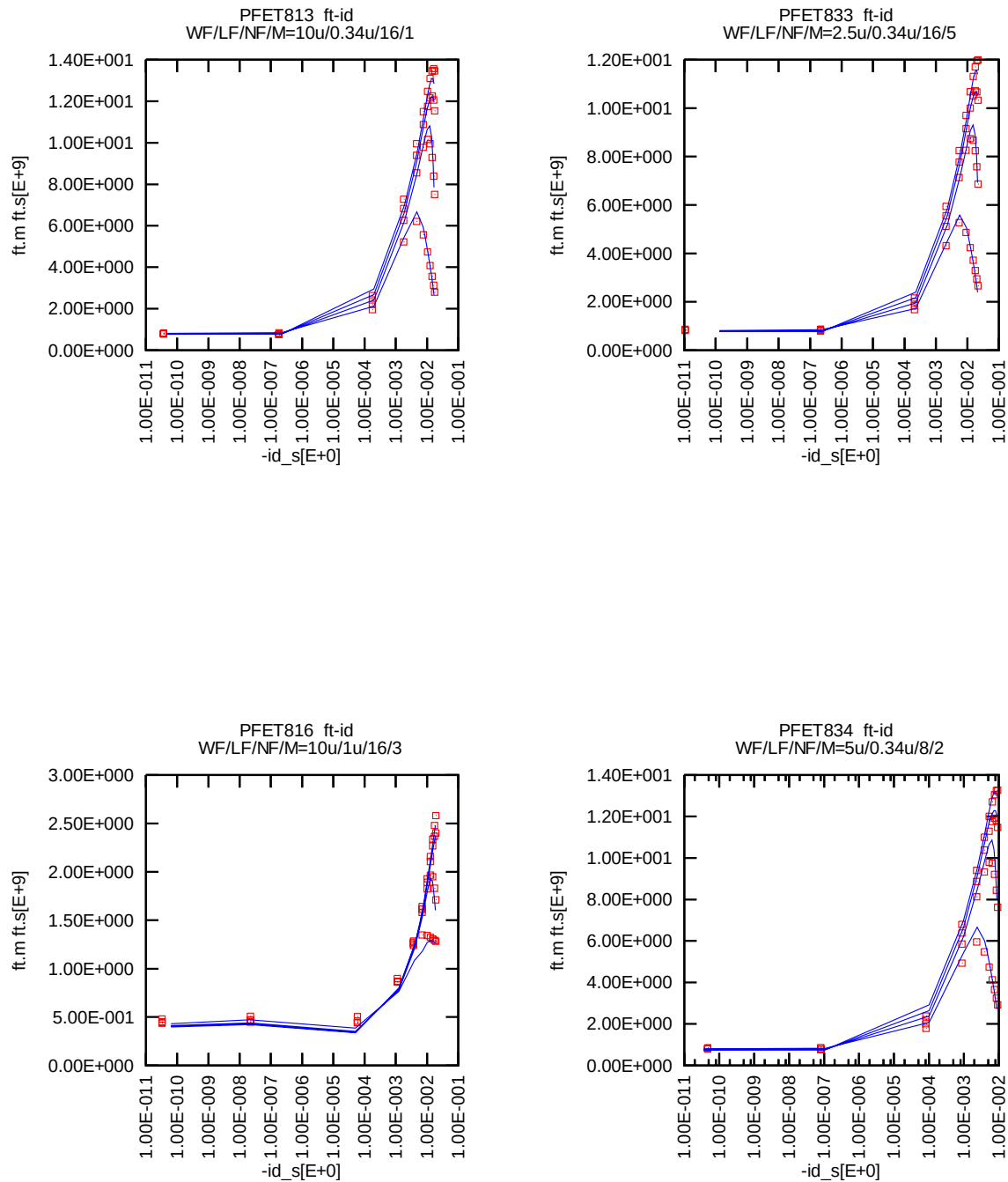


Figure 2-24 f_T -Id (25°C)

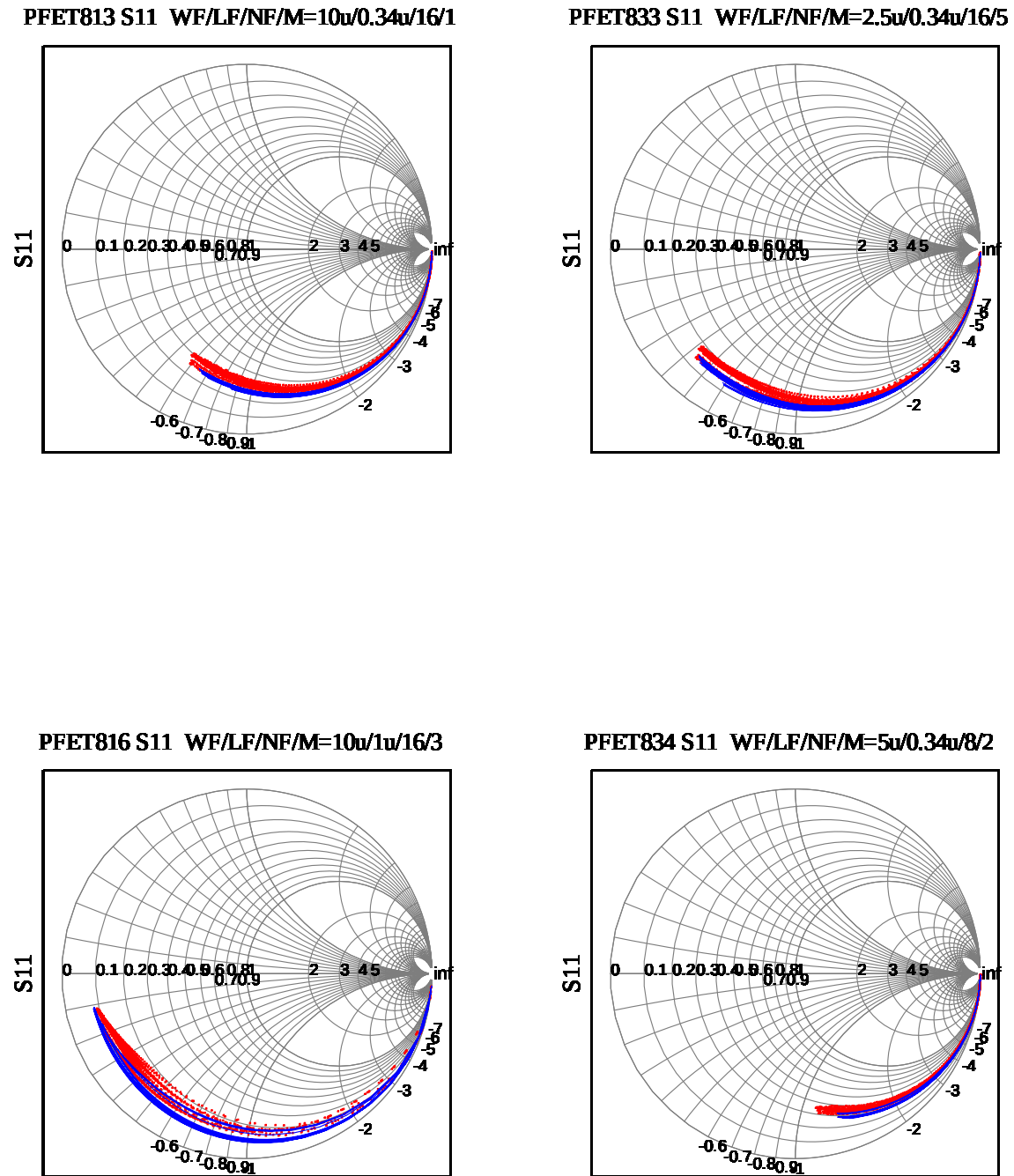


Figure 2-25 S11 (25°C)

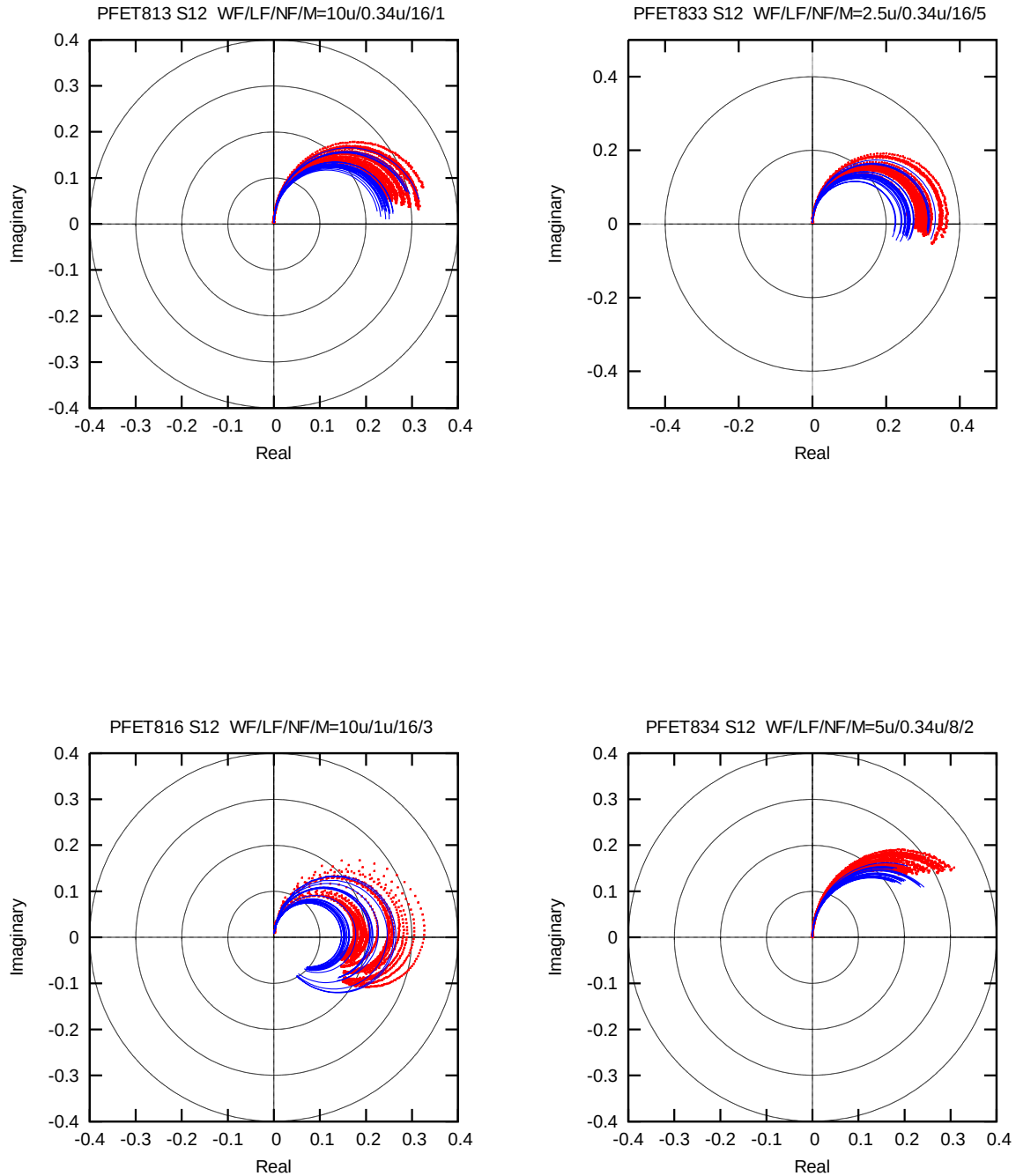


Figure 2-26 S12 (25°C)

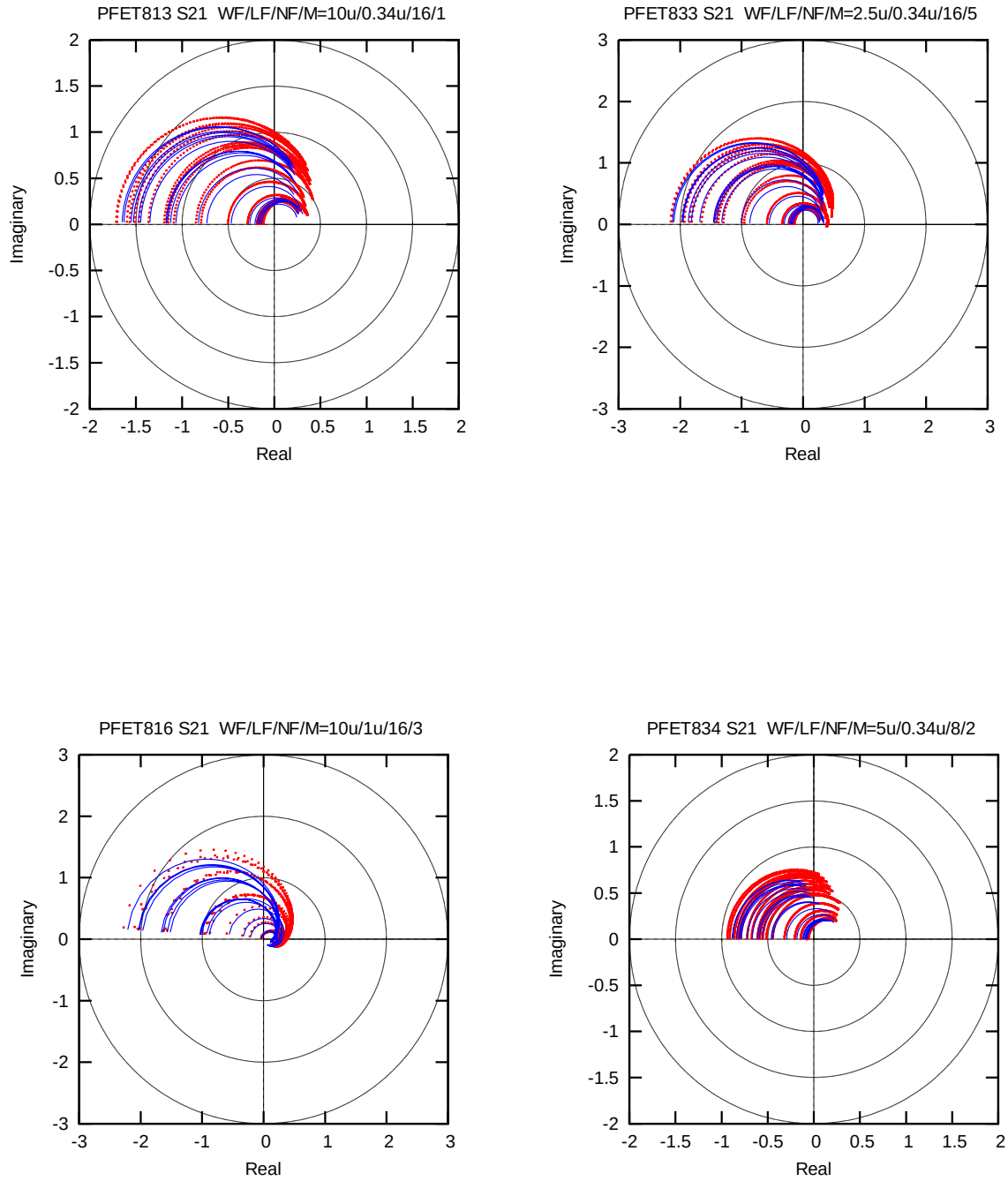


Figure 2-27 S21 (25°C)

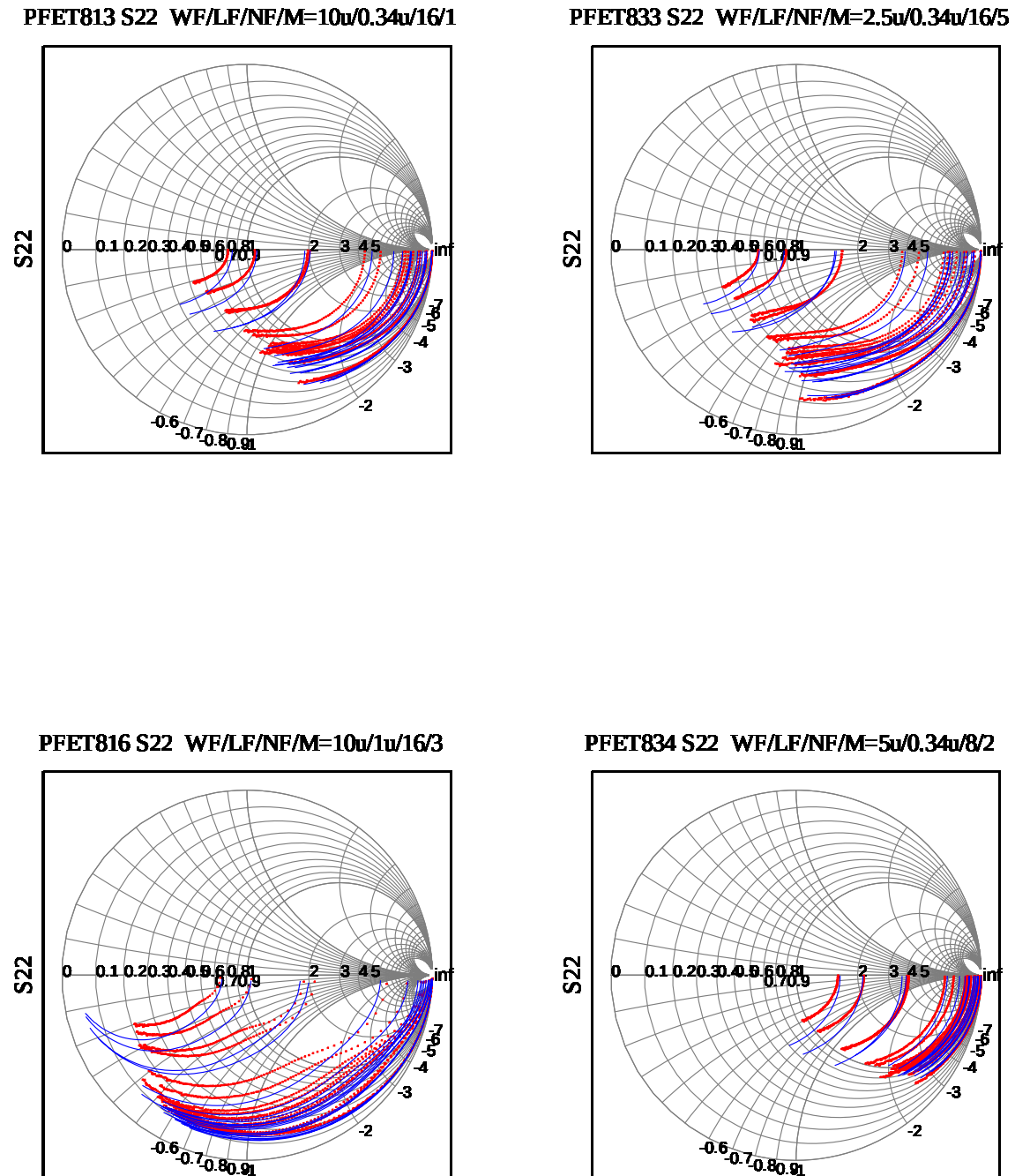


Figure 2-28 S22 (25°C)

3 High Frequency Noise Model Enhancement

3.1 General Information

This section describes the MOSFET RF models with High Frequency (HF) noise enhancement. At the current release, all four simulators (**Spectre, Hspice and Eldo**) models support HF noise model enhancement.

Except noise behavior, the RF models with and without HF noise enhancement have exact same electrical behaviors.

However, the simulation with the HF noise enhancement generally needs more simulation time than the simulation without HF noise enhancement since Verilog-A code is used in the HF noise enhancement. It is recommended to use the model without HF noise enhancement for simulations other than noise analysis, and use the RF model with HF noise enhancement for noise analysis to get more accurate noise results.

3.2 Model Usage

3.2.1 Voltage Scaling Range

The following voltage scaling range has been used to extract and verify high-frequency noise parameters:

Vg: 0.9 ~ 3.3V

Vd: 1 ~ 3.3V

Vs: 0V

Vb: 0V

If the device is biased outside the above voltage range, the HF noise enhancement may not accurately predict the noise performance.

3.2.2 Global and Local Flag Parameters for HF Noise Simulation

Two levels of flag parameters are designed to switch on or off the HF noise enhancement:

hfnoise (global flag): 1 is on and 0 is off

“hfnoise” need to be pre-defined. If using netlist, add the following line in the netlist:
parameters hfnoise=1 (Spectre)

.param hfnoise=1 (Hspice and Eldo)

If using Cadence Analog Environment, “hfnoise” need to be defined as a Design Variable before simulation.

hfn_flag (local flag): 1 is on and 0 is off

“hfn_flag” is a model input parameter. If it is not given in the netlist, its default value will equal to the global parameter “hfnoise”

The following examples demonstrate how to use these two flag parameters:

Example 1: HF noise enhancement switched off for transistors x1, x2, x3

Parameters hfnoise=0

x1 (d1 g1 s1 b1) n_33_rf lf=0.34u wf=12u nf=16 m=1

x2 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1

x3 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1

Example 2: HF noise enhancement switched on for transistors x1, x2, x3

Parameters hfnoise=1

x1 (d1 g1 s1 b1) n_33_rf lf=0.34u wf=12u nf=16 m=1

x2 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1

x3 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1

Example 3: HF noise enhancement switched off for transistors x2, x3 but switched on for x1

Parameters *hfnoise=0*

x1 (d1 g1 s1 b1) n_33_rf lf=0.34u wf=12u nf=16 m=1 hfn_flag=1

x2 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1

x3 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1 hfn_flag=0

Example 4: HF noise enhancement switched on for transistors x1, x2, but switched off for x3

Parameters *hfnoise=1*

x1 (d1 g1 s1 b1) n_33_rf lf=0.34u wf=12u nf=16 m=1 hfn_flag=1

x2 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1

x3 (d2 g2 s2 b2) n_33_rf lf=0.34u wf=12u nf=16 m=1 hfn_flag=0

3.2.3 Multiple Factor “mf” for Hspice and Eldo

Due to the fact that Verilog-A codes in Hspice and Eldo don’t support multiple factor which is inherited from the netlist, an additional model input parameter “*mf*” has to be defined in order to accurately simulate the High Frequency noise for multiple devices connected in parallel. *mf* will not affect the simulation results other than thermal noise behavior. The value of *mf* should be always be equal to the value of instant multiple factor parameter *m*. Please refer to the following example:

If 4 devices are connected in parallel, the netlist for Hspice and Eldo should be:

x1 (d g s b) n_33_rf lf=0.34u wf=12u nf=16 **m=1 mf=1**

3.3 HF Noise Modeling Result

3.3.1 NMOS Noise Parameters – NF_{min} , R_n & G_{opt}

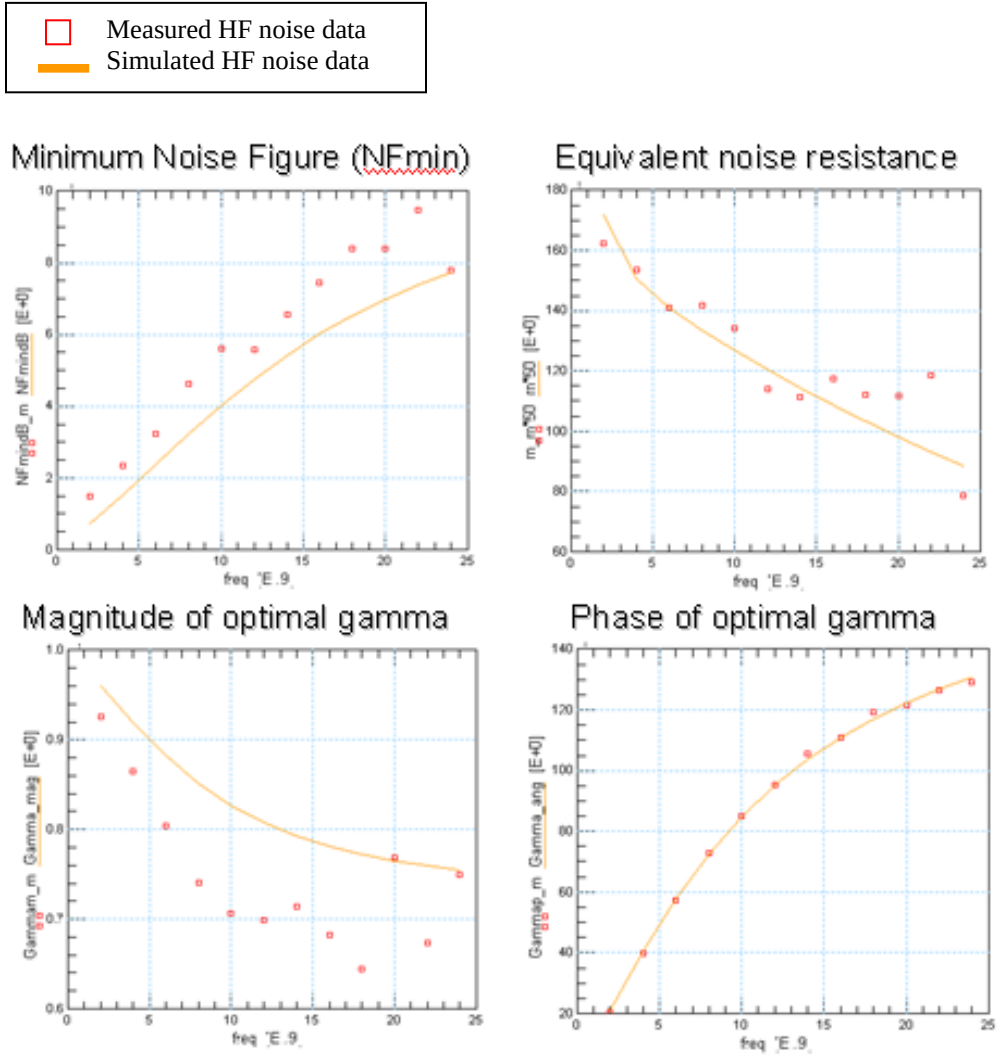


Figure 3-1 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.34um, WF=12um, NF=16, m=1 @ Vd=1V, Vg=2.5V

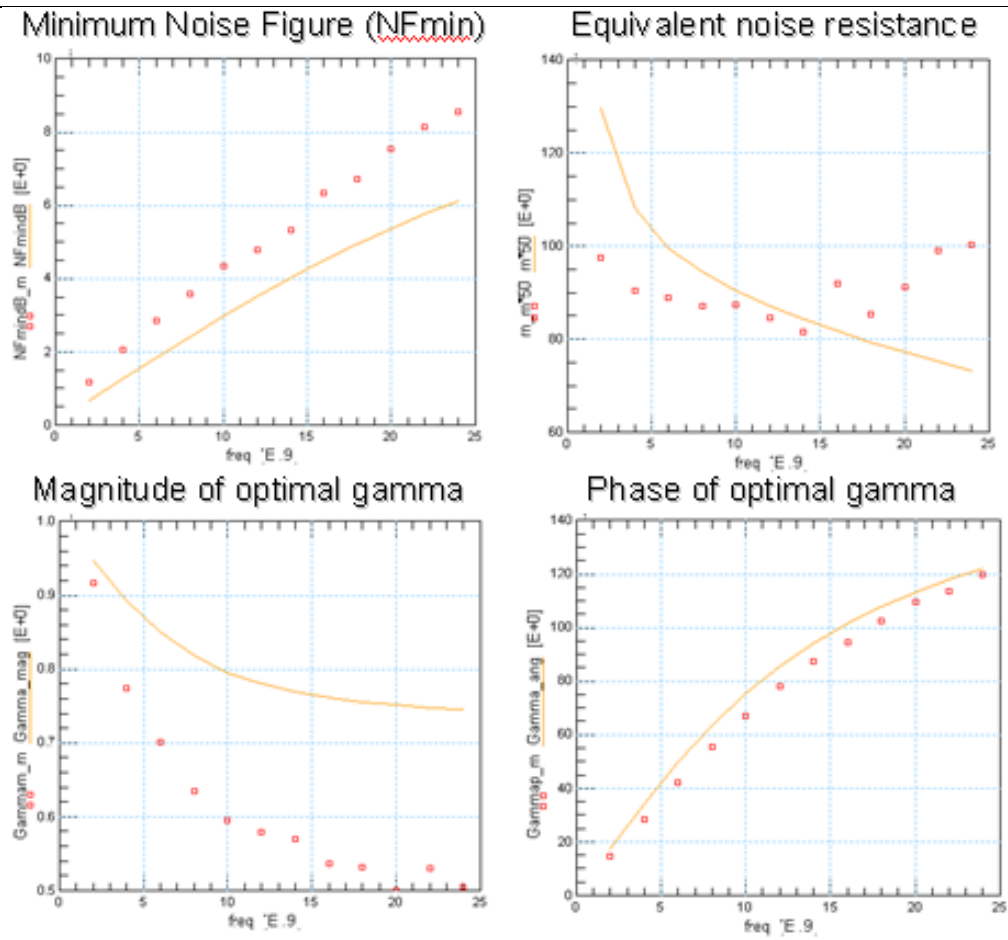
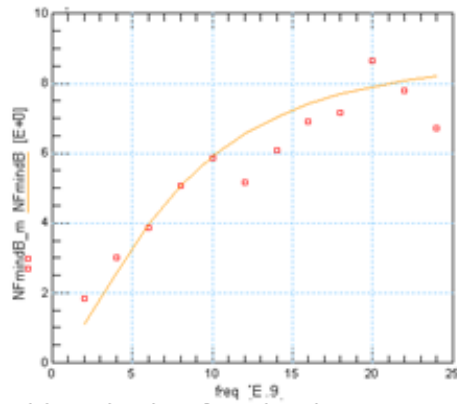
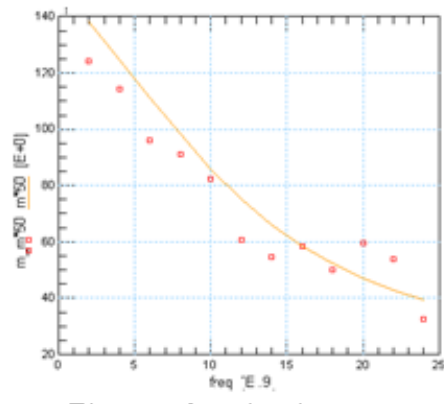


Figure 3-2 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.34 μ m, WF=12 μ m, NF=16, m=1 @ Vd=3.3V, Vg=3.3V

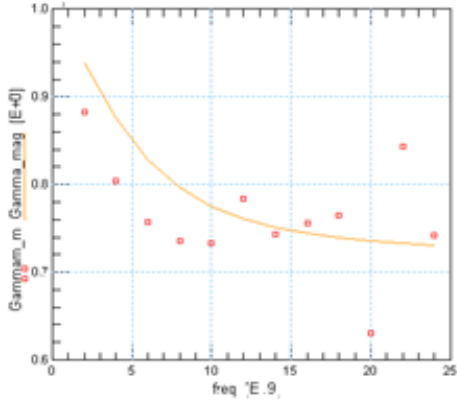
Minimum Noise Figure (NFmin)



Equivalent noise resistance



Magnitude of optimal gamma



Phase of optimal gamma

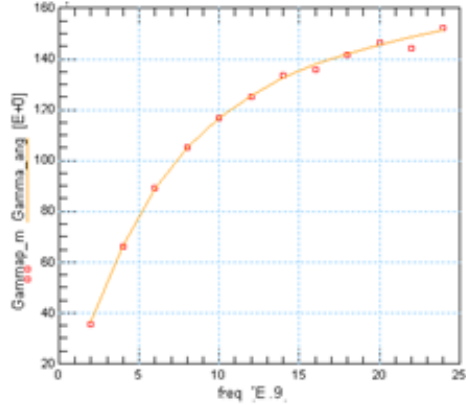


Figure 3-3 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.7um, WF=12um, NF=16, m=1 @ Vd=1V, Vg=2.5V

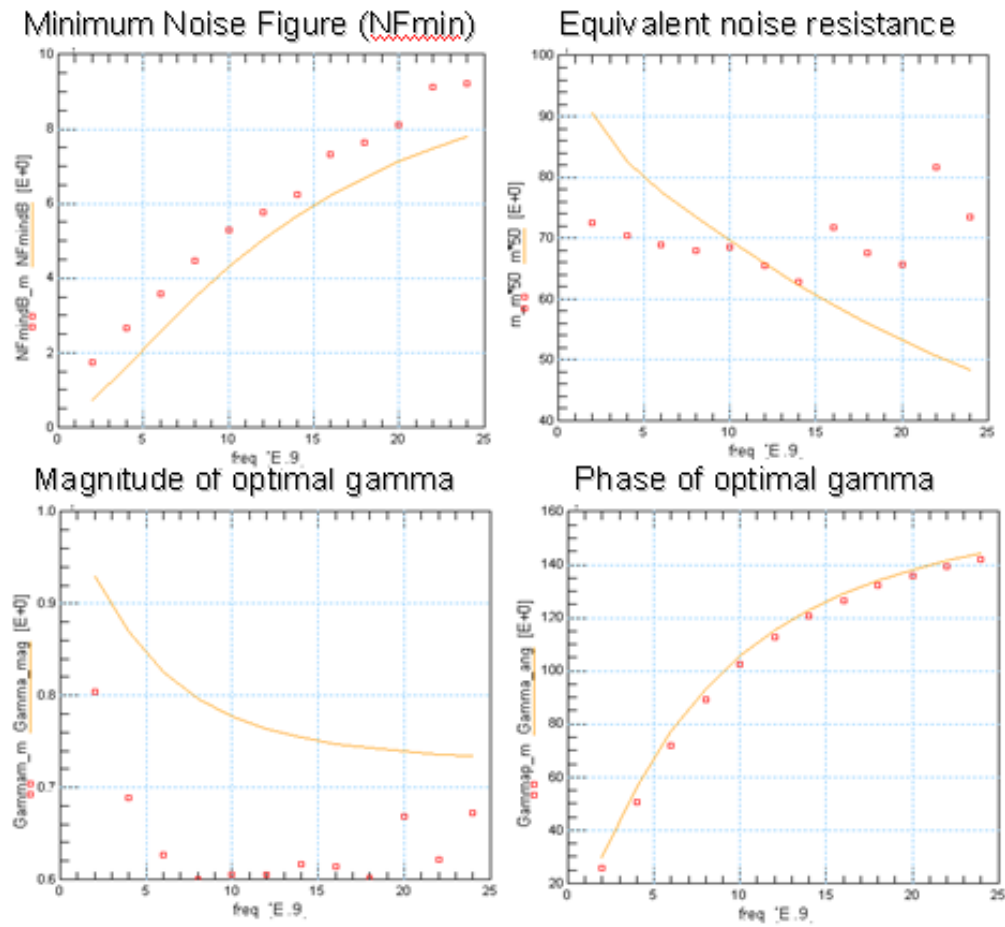


Figure 3-4 Measured and simulated noise parameters vs. Frequency for NMOS: LF=0.7μm, WF=12μm, NF=16, m=1 @ Vd=3.3V, Vg=3.3V

3.3.2 PMOS Noise Parameters – NF_{min} , R_n & G_{opt}

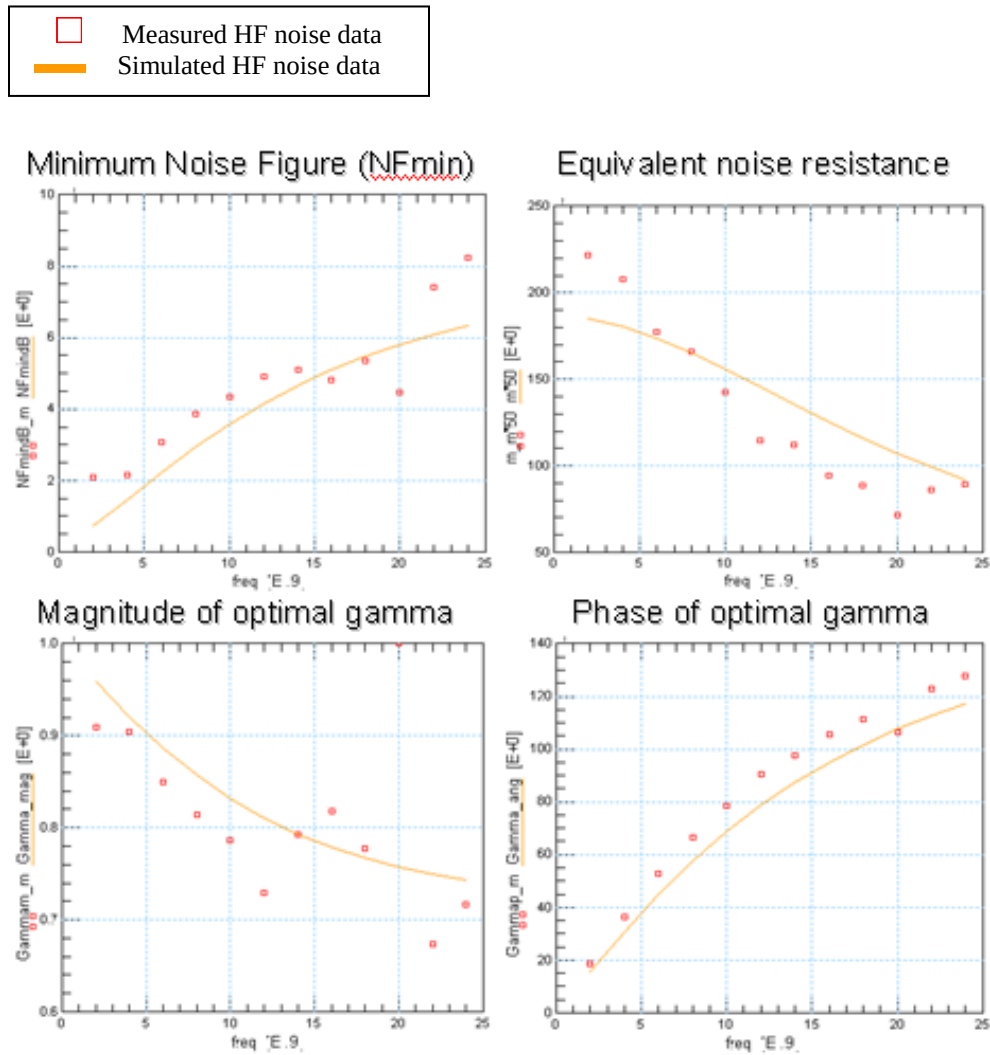
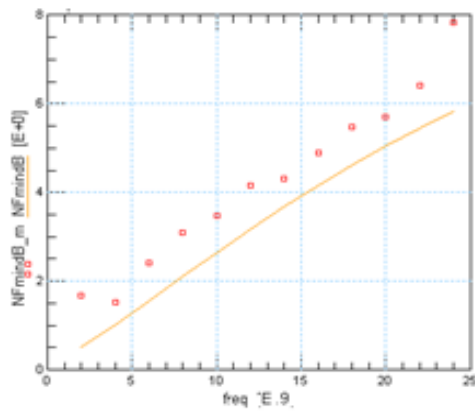
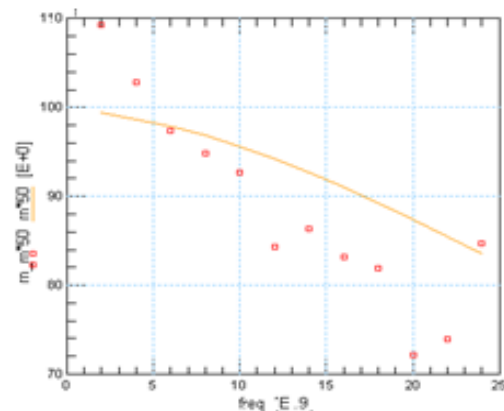


Figure 3-5 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.34um, WF=10um, NF=16, m=1 @ $V_d=-1V$, $V_g=-2.5V$

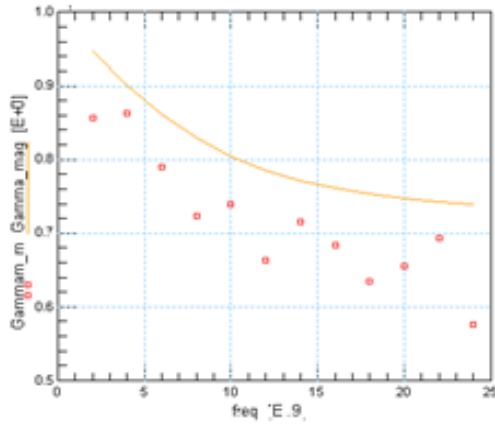
Minimum Noise Figure (NFmin)



Equivalent noise resistance



Magnitude of optimal gamma



Phase of optimal gamma

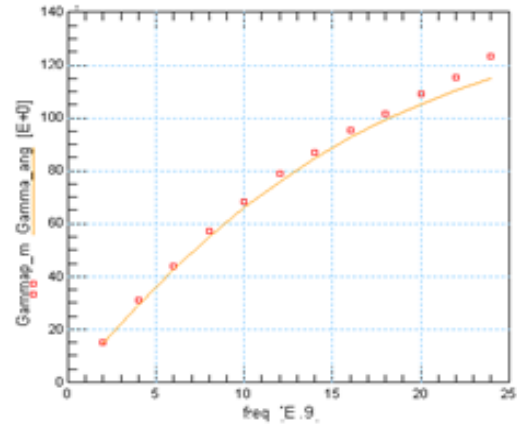


Figure 3-6 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.34um, WF=10um, NF=16, m=1 @ Vd=-3.3V, Vg=-3.3V

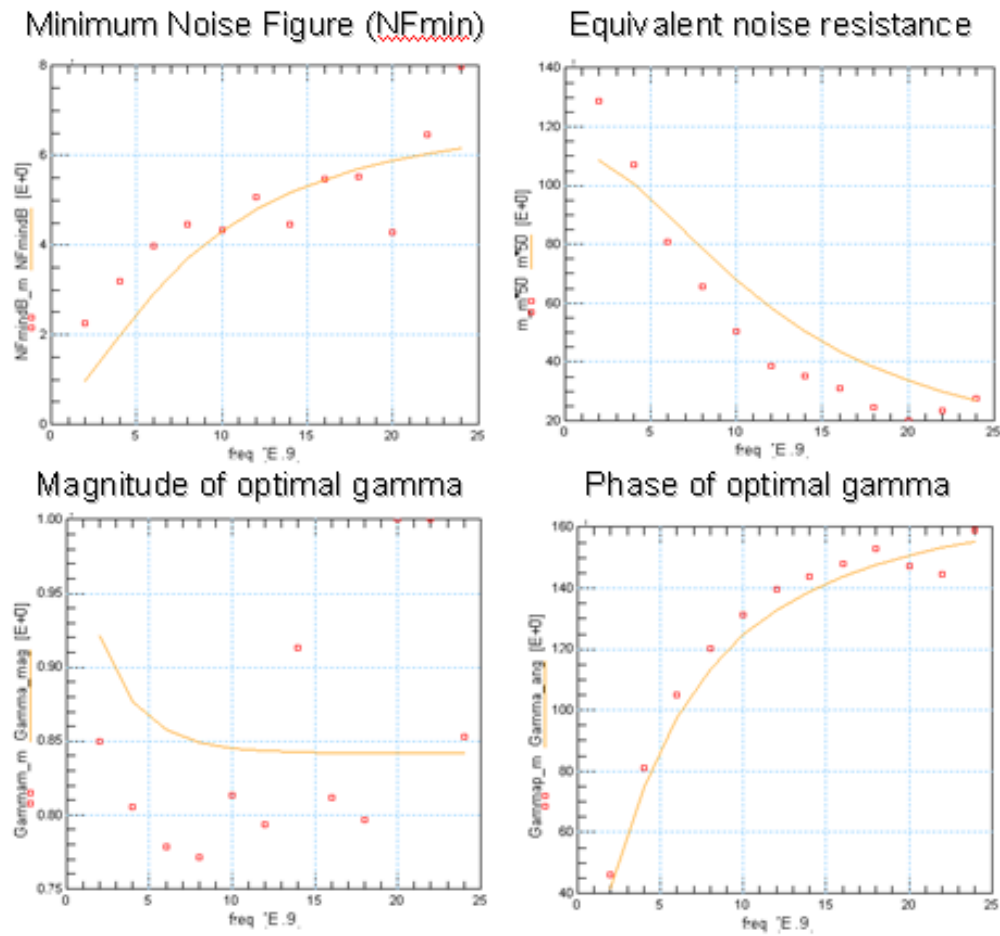


Figure 3-7 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.5um, WF=10um, NF=16, m=2 @ Vd=-1V, Vg=-2.5V

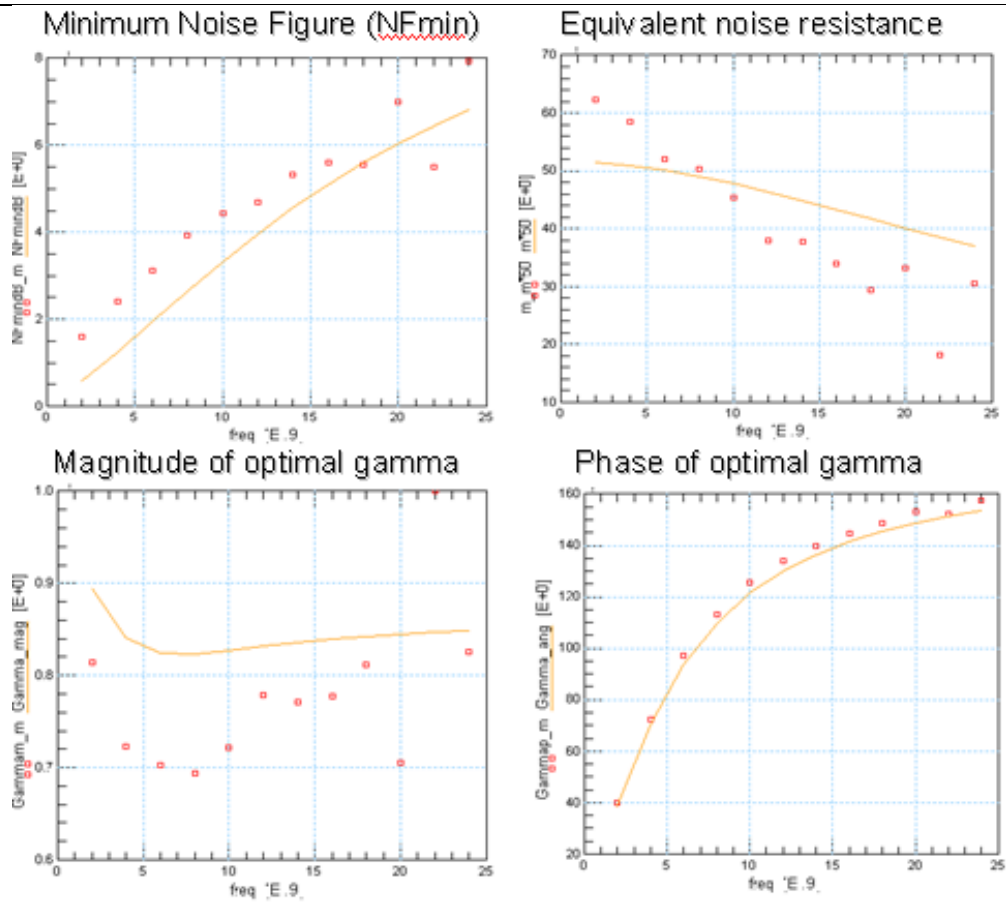


Figure 3-8 Measured and simulated noise parameters vs. Frequency for PMOS: LF=0.5um, WF=10um, NF=16, m=2 @ Vd=-2.5V, Vg=-3.3V

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	N/A	

4.2 SPECTRE Warning Message

Table 4-2 SPECTRE warning message

NO.	Warning Message	Explanation
1.	N/A	

4.3 ELDO Warning Message

Table 4-3 ELDO warning message

NO.	Warning Message	Explanation
1.	N/A	